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Zhang et al.

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(45) **Date of Patent:** **Aug. 26, 2025**

(54) **CHIP PACKAGE ON PACKAGE
STRUCTURE, PACKAGING METHOD
THEREOF, AND ELECTRONIC DEVICE**

(71) Applicant: **Huawei Technologies Co., Ltd.**,
Shenzhen (CN)

(72) Inventors: **Tonglong Zhang**, Shenzhen (CN);
Xiaodong Zhang, Shenzhen (CN);
Yong Guan, Shanghai (CN); **Simin
Wang**, Shanghai (CN)

(73) Assignee: **HUAWEI TECHNOLOGIES CO.,
LTD.**, Shenzhen (CN)

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(51) **Int. Cl.**
H01L 23/00 (2006.01)
H01L 25/10 (2006.01)

(52) **U.S. Cl.**
CPC **H01L 24/08** (2013.01); **H01L 24/09**
(2013.01); **H01L 24/80** (2013.01); **H01L 24/81**
(2013.01);
(Continued)

(58) **Field of Classification Search**
CPC H01L 23/488; H01L 24/05; H01L 24/08;
H01L 24/09; H01L 24/16; H01L 24/17;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

2014/0035162 A1* 2/2014 Khan H01L 23/49816
257/774

2014/0103488 A1 4/2014 Chen et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CN 103915421 A 7/2014
CN 104505382 A 4/2015

(Continued)

OTHER PUBLICATIONS

Zhang, C., et al., "Thermo-structural model of stacked field-
programmable gate arrays with through-silicon vias", Electronics
Letters, vol. 45(24), pp. 1236-1238 (Year: 2009).*

Primary Examiner — Fernando L Toledo

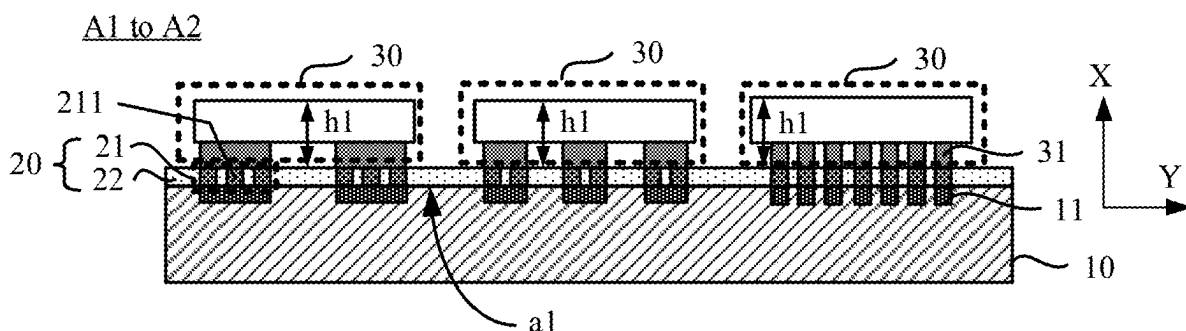
Assistant Examiner — Ryan T. Fortin

(74) *Attorney, Agent, or Firm* — Conley Rose, P.C.

(57) **ABSTRACT**

A chip package on package structure includes a primary chip
stack unit having pins insulated and spaced from each other
on a first surface; a first bonding layer disposed on the first
surface, where the first bonding layer includes bonding
components insulated and spaced from each other, each
bonding component includes a bonding part, and any two
bonding parts are insulated and have a same cross-sectional
area, and the bonding components are separately bonded to
the pins; and secondary chip stack units, disposed on a
surface of a side that is of the first bonding layer and that is
away from the primary chip stack unit, where the secondary
chip stack unit has micro bumps insulated and spaced from
each other, and each of the micro bumps is bonded to one of
the bonding components.

20 Claims, 22 Drawing Sheets



(52) **U.S. Cl.**

CPC **H01L 25/105** (2013.01); *H01L 24/05* (2013.01); *H01L 24/16* (2013.01); *H01L 24/17* (2013.01); *H01L 24/32* (2013.01); *H01L 24/33* (2013.01); *H01L 24/73* (2013.01); *H01L 24/96* (2013.01); *H01L 24/97* (2013.01); *H01L 2224/05647* (2013.01); *H01L 2224/0801* (2013.01); *H01L 2224/08057* (2013.01); *H01L 2224/08148* (2013.01); *H01L 2224/0903* (2013.01); *H01L 2224/09051* (2013.01); *H01L 2224/16145* (2013.01); *H01L 2224/1703* (2013.01); *H01L 2224/17051* (2013.01); *H01L 2224/32145* (2013.01); *H01L 2224/3303* (2013.01); *H01L 2224/33051* (2013.01); *H01L 2224/73204* (2013.01); *H01L 2224/80203* (2013.01); *H01L 2224/80805* (2013.01); *H01L 2224/80895* (2013.01); *H01L 2224/80896* (2013.01); *H01L 2224/81203* (2013.01); *H01L 2224/81805* (2013.01); *H01L 2224/81815* (2013.01); *H01L 2225/1023* (2013.01); *H01L 2225/1041* (2013.01); *H01L 2225/107* (2013.01); *H01L 2225/1094* (2013.01)

(58) **Field of Classification Search**

CPC H01L 24/19; H01L 24/32; H01L 24/33; H01L 24/73; H01L 24/80; H01L 24/81; H01L 24/96; H01L 24/97; H01L 25/105; H01L 25/03; H01L 25/0652; H01L 25/50; H01L 25/0655; H01L 2224/05647; H01L 2224/0801; H01L 2224/08057; H01L

2224/08148; H01L 2224/0903; H01L 2224/09051; H01L 2224/16145; H01L 2224/1703; H01L 2224/17051; H01L 2224/32145; H01L 2224/3303; H01L 2224/33051; H01L 2224/73204; H01L 2224/80203; H01L 2224/80805; H01L 2224/80895; H01L 2224/80896; H01L 2224/81203; H01L 2224/81805; H01L 2224/81815; H01L 2225/1023; H01L 2225/1041; H01L 2225/107; H01L 2225/1094

See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

2014/0185264 A1 7/2014 Chen et al.
 2014/0312490 A1* 10/2014 Yang H01L 25/03
 257/737
 2015/0214192 A1* 7/2015 Lin H01L 23/49816
 257/737
 2016/0329262 A1* 11/2016 Hsiao H01L 23/49894
 2018/0076179 A1* 3/2018 Hsu H01L 21/56
 2019/0244905 A1* 8/2019 Yu H01L 23/5383
 2020/0091128 A1* 3/2020 Elsherbini H01L 23/5386

FOREIGN PATENT DOCUMENTS

CN 107785339 A 3/2018
 KR 10-2015-0084287 * 7/2015 H01L 23/28

* cited by examiner

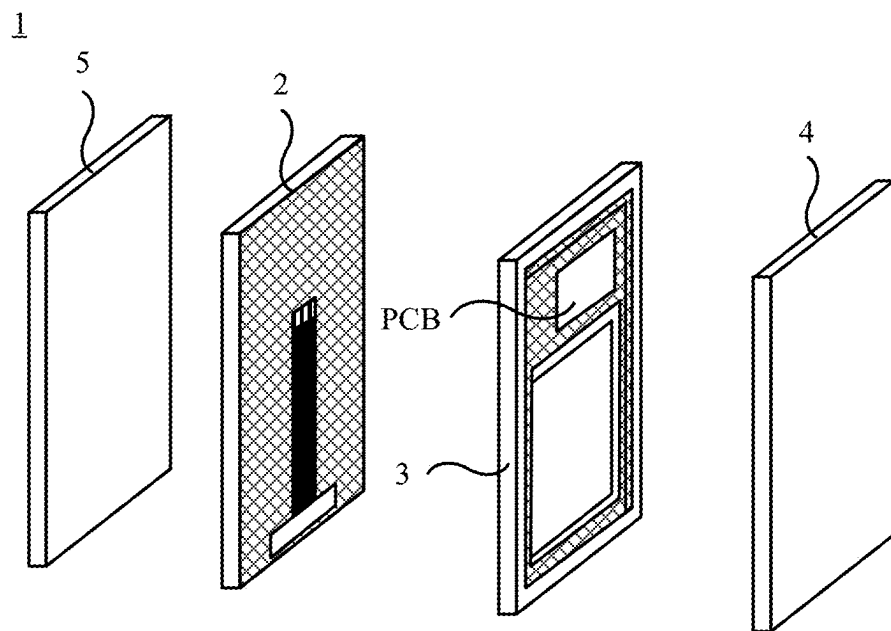


FIG. 1

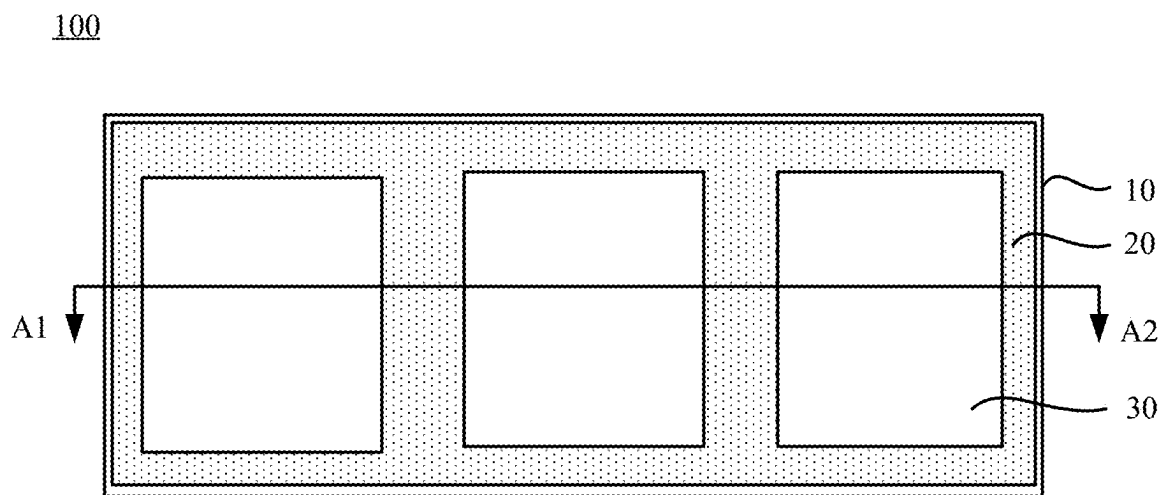


FIG. 2A

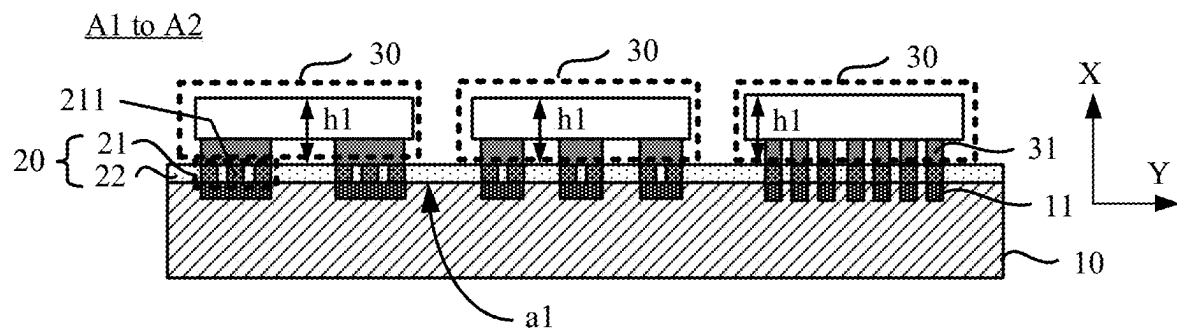


FIG. 2B

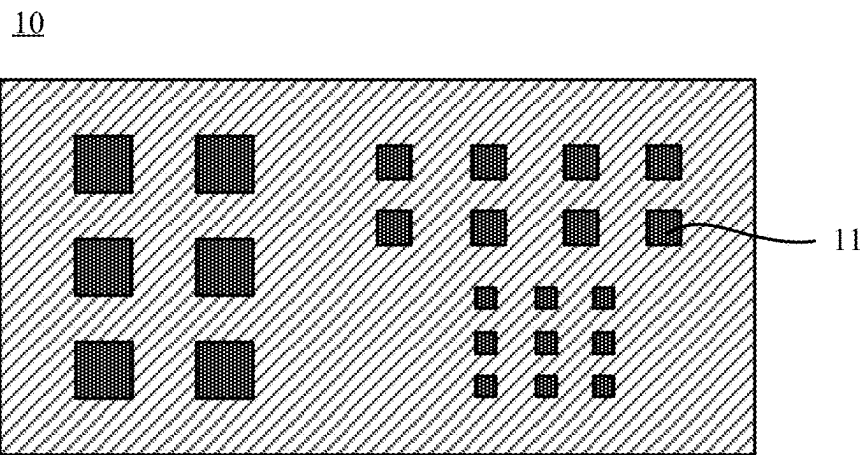


FIG. 2C

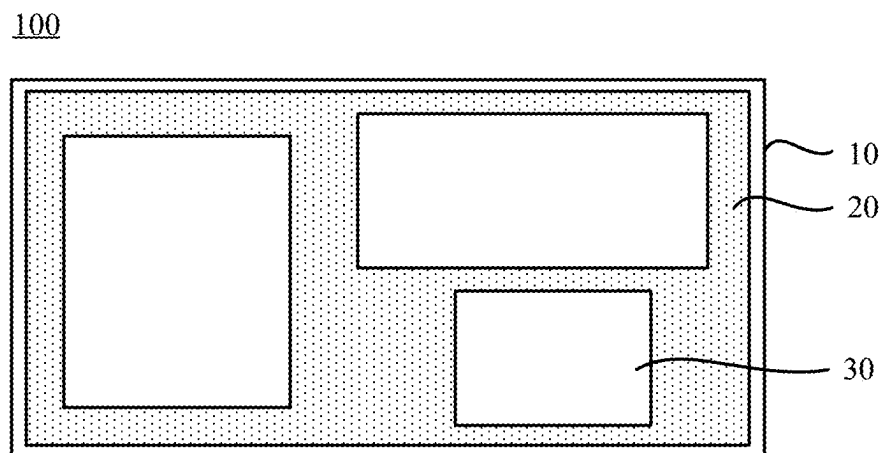


FIG. 2D

A1 to A2

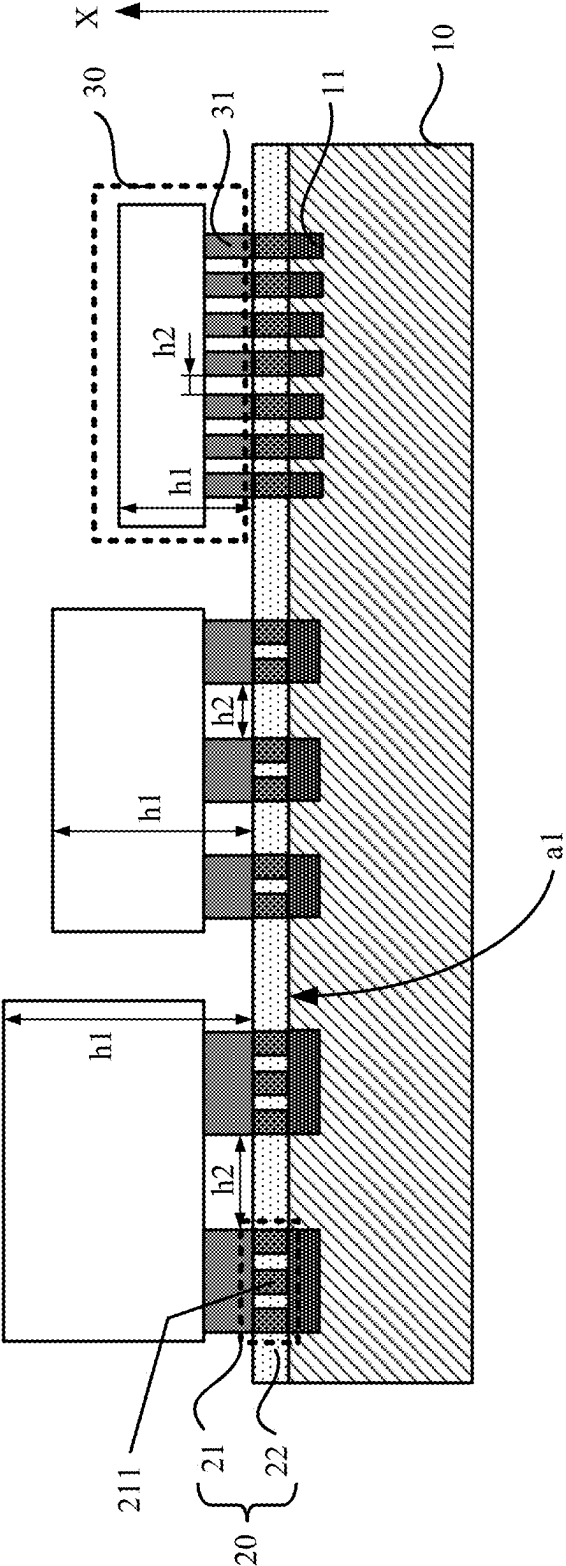


FIG. 2E

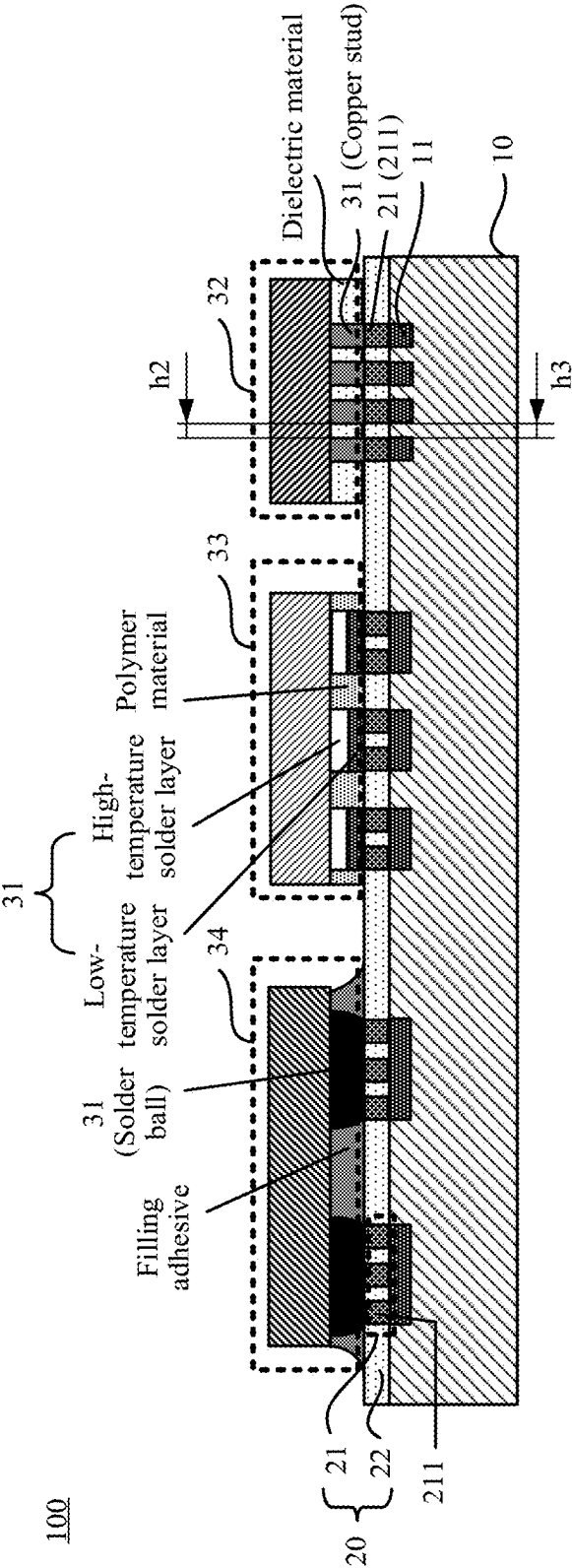


FIG. 3A

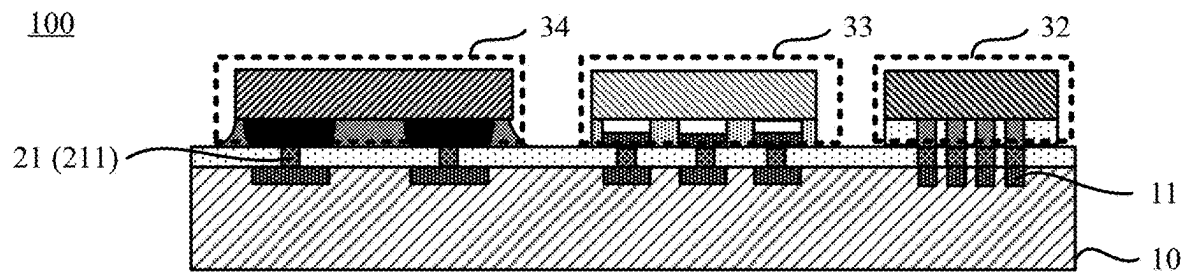


FIG. 3B

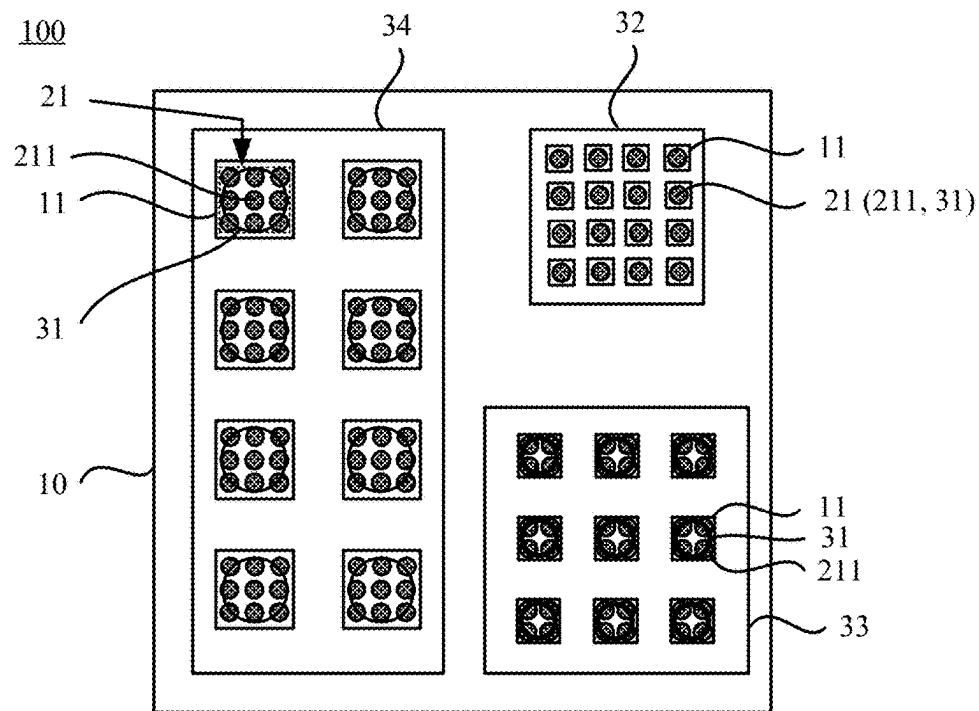


FIG. 3C

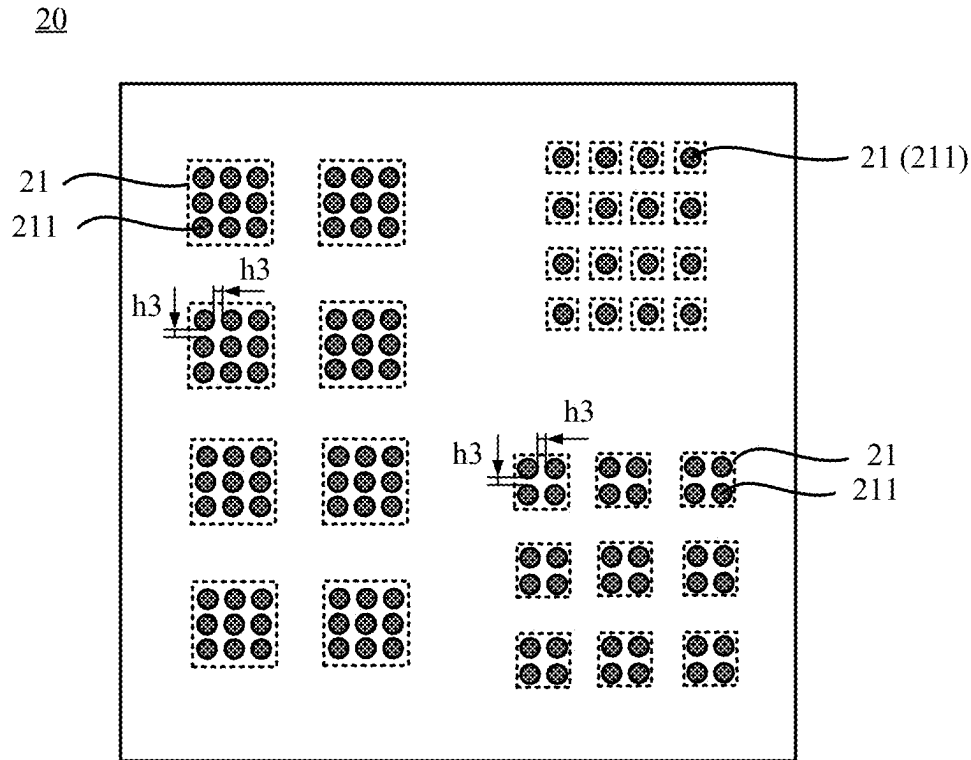


FIG. 4

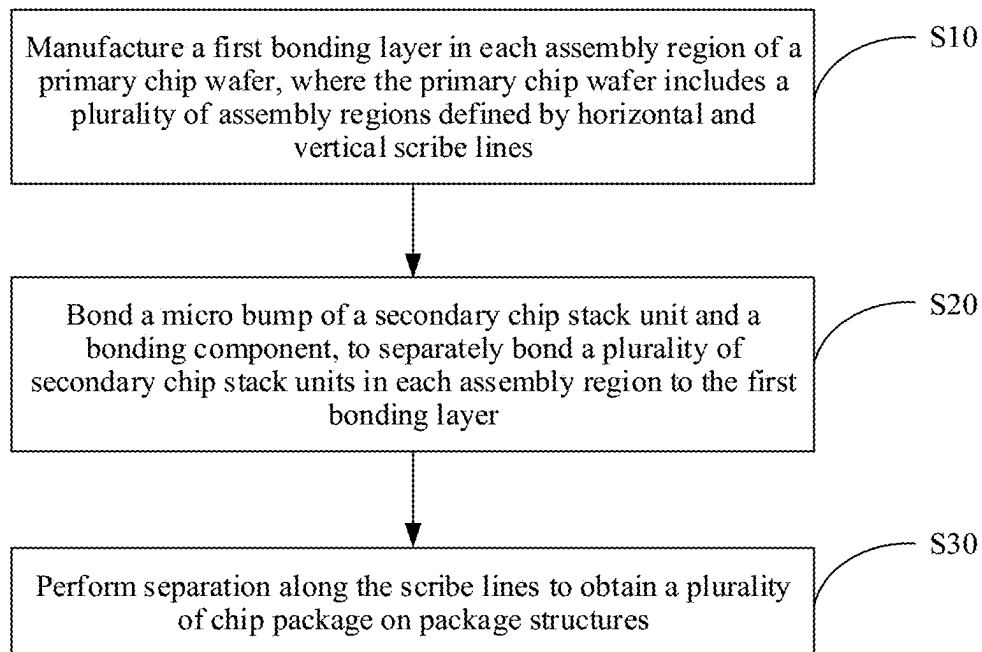


FIG. 5A

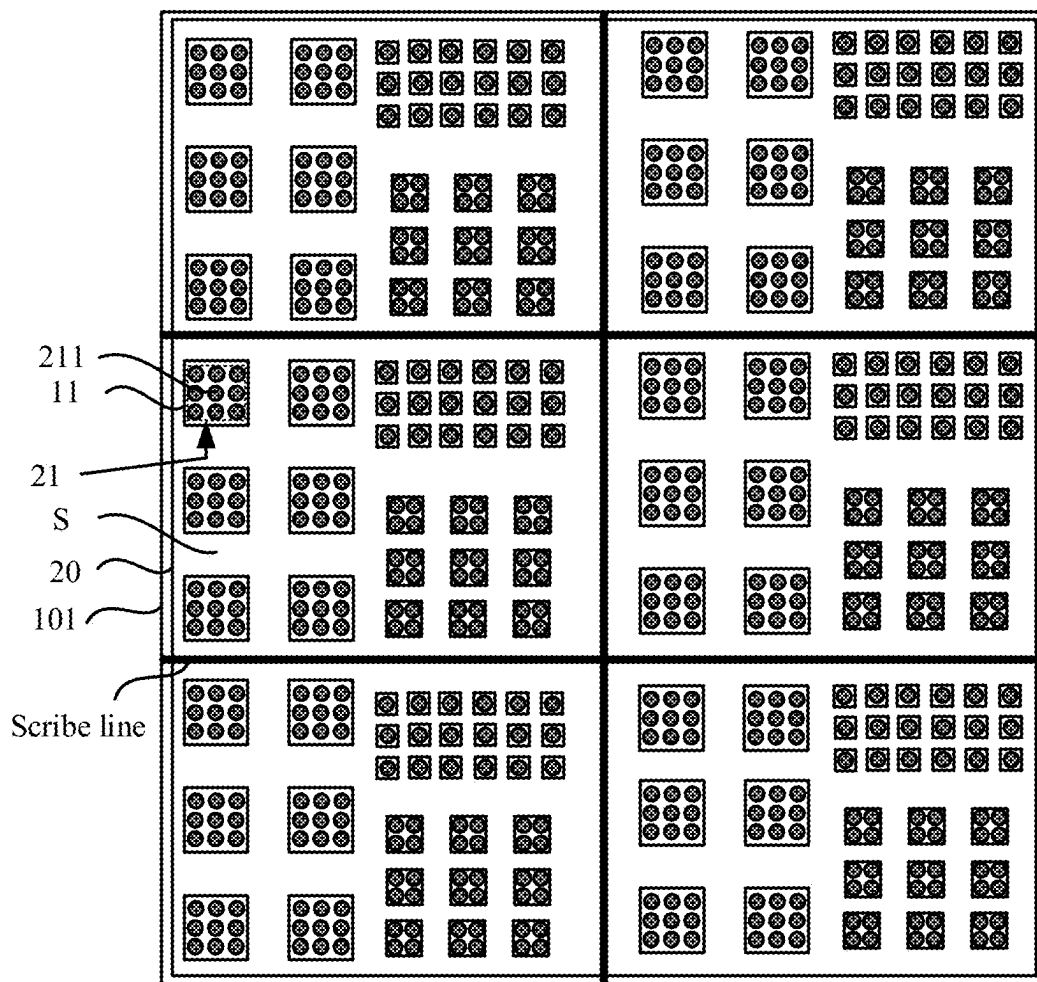


FIG. 5B

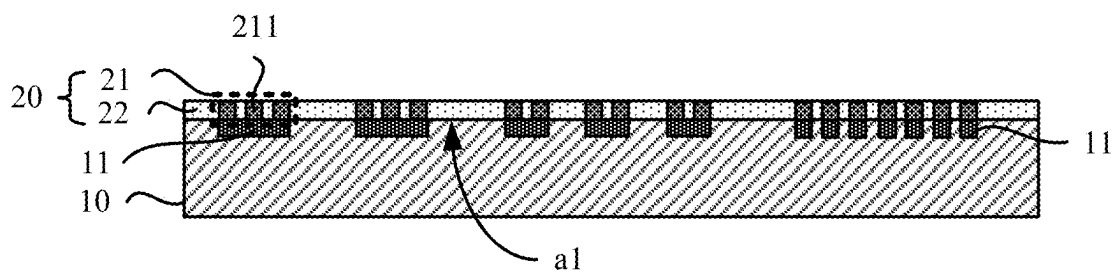


FIG. 5C

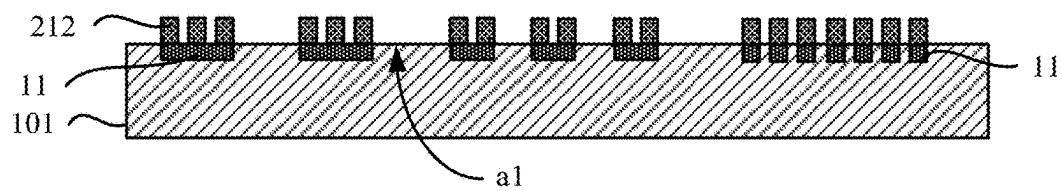


FIG. 5D

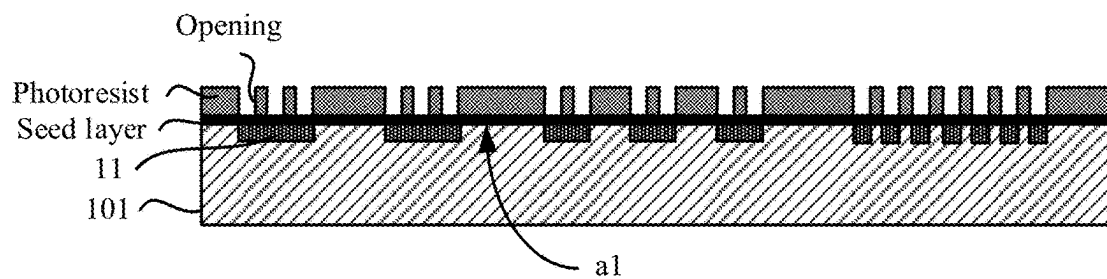


FIG. 5E

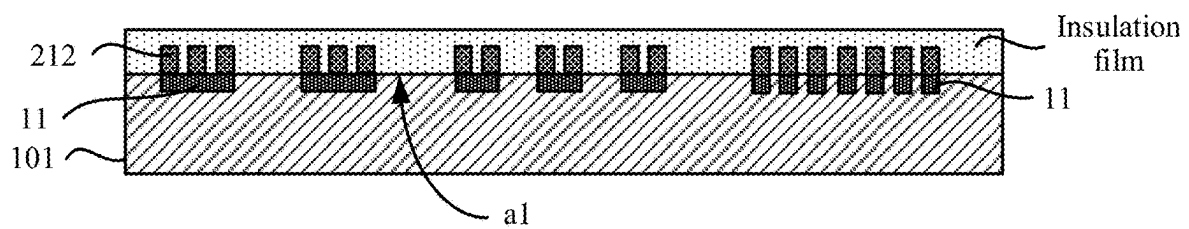


FIG. 5F

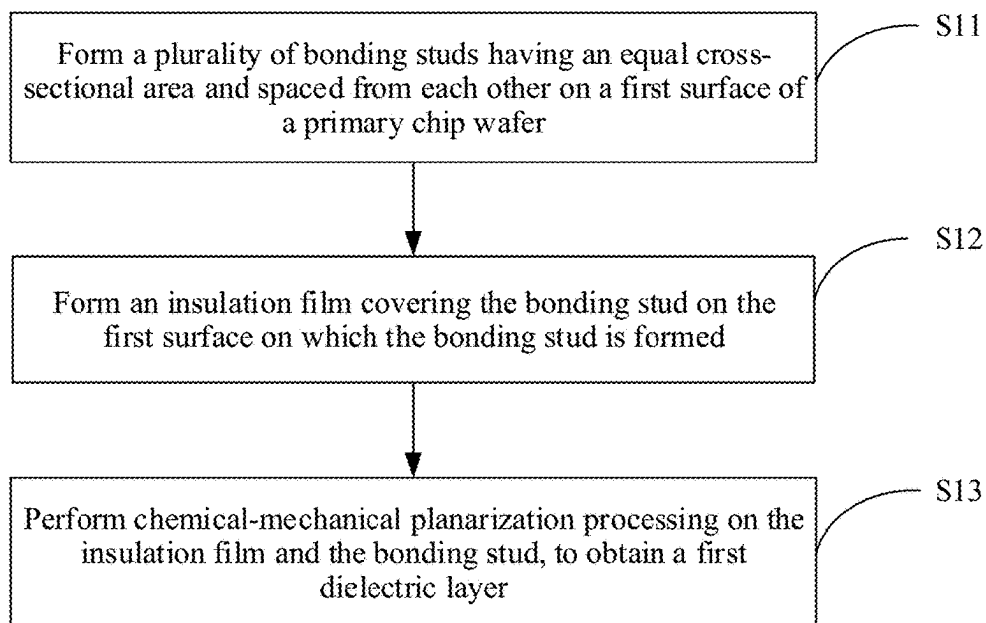


FIG. 5G

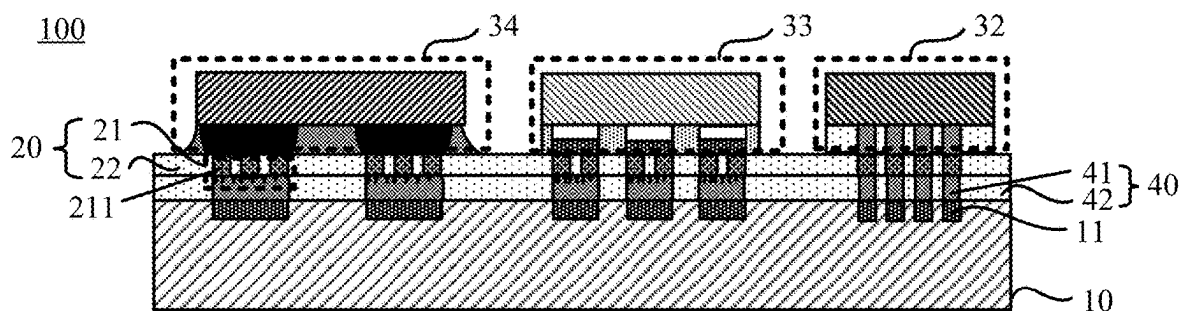


FIG. 6A

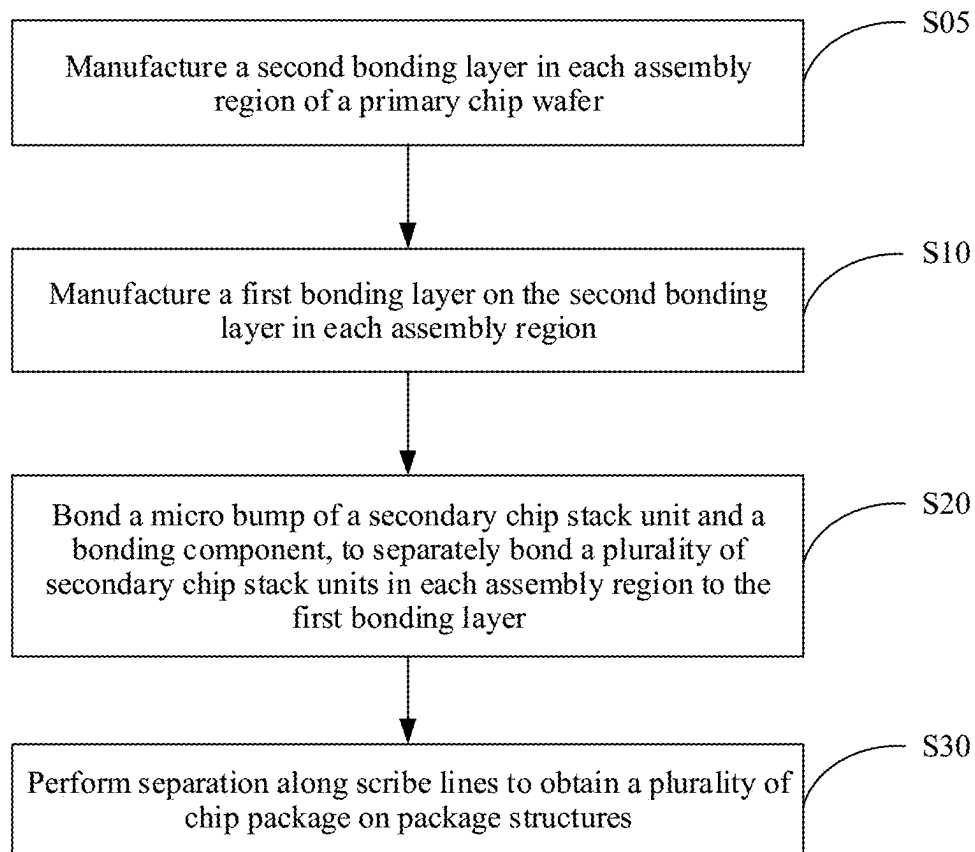


FIG. 6B

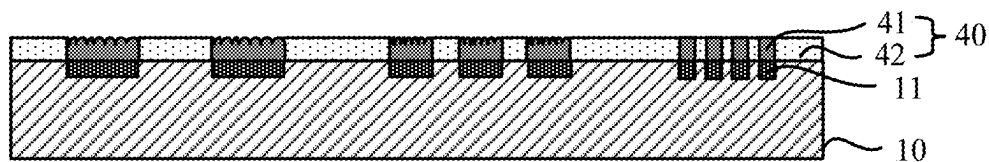


FIG. 6C

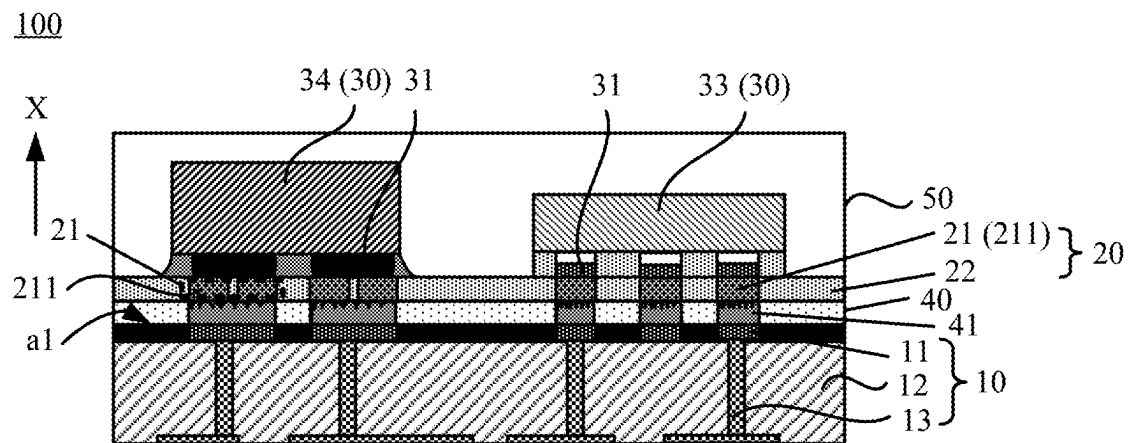


FIG. 7A

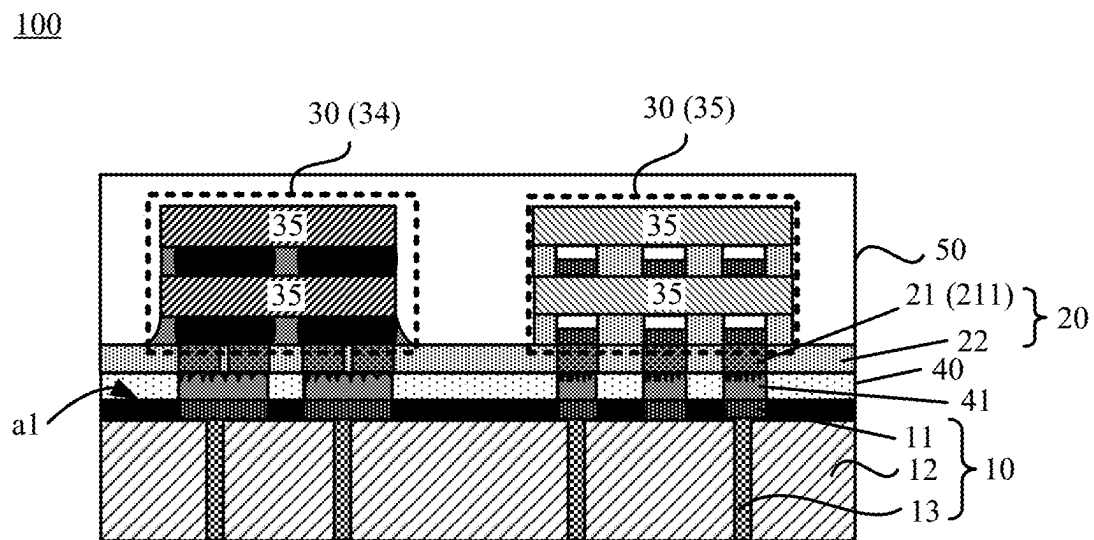


FIG. 7B

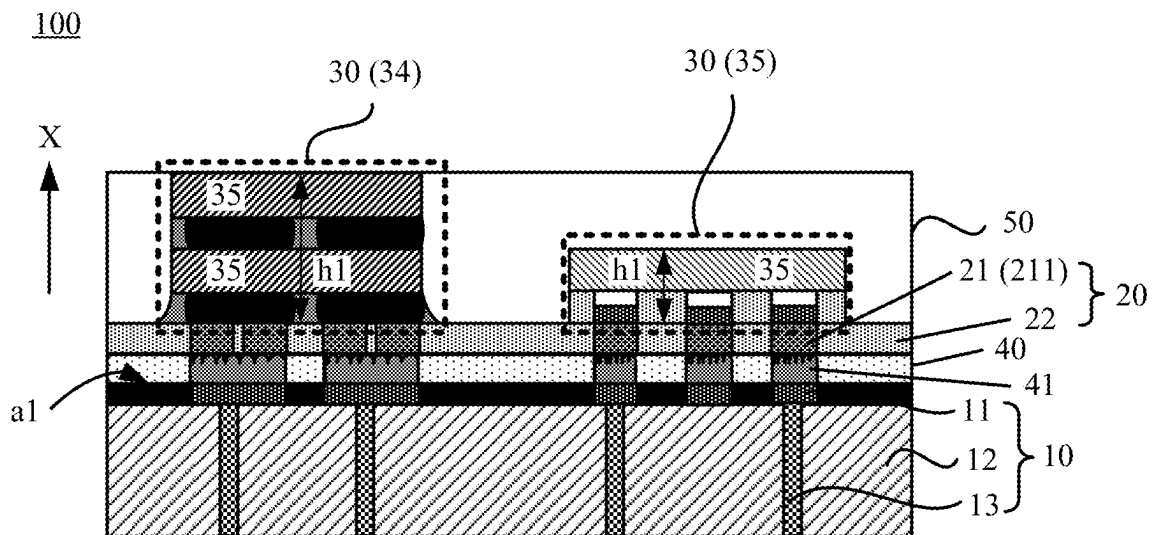


FIG. 7C

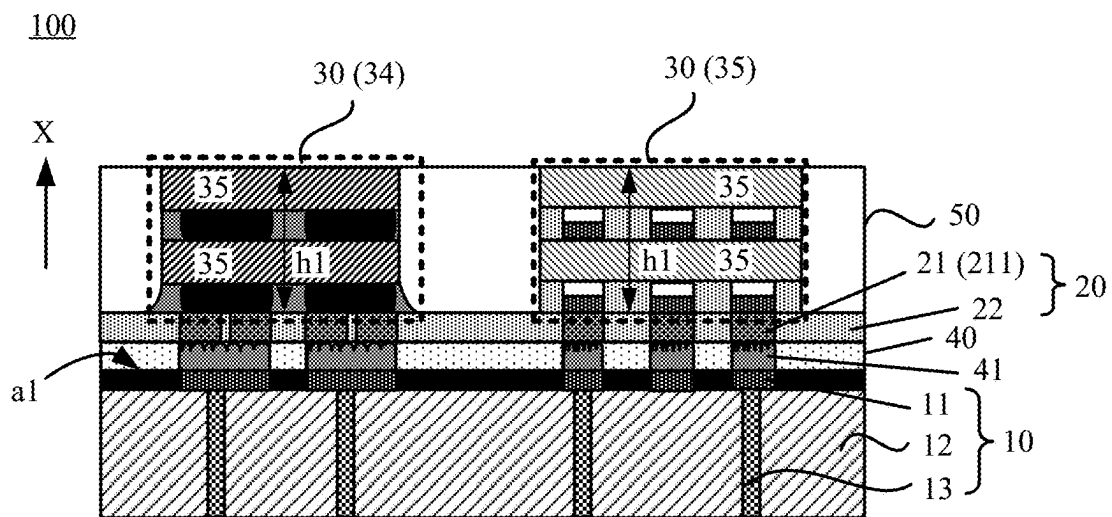
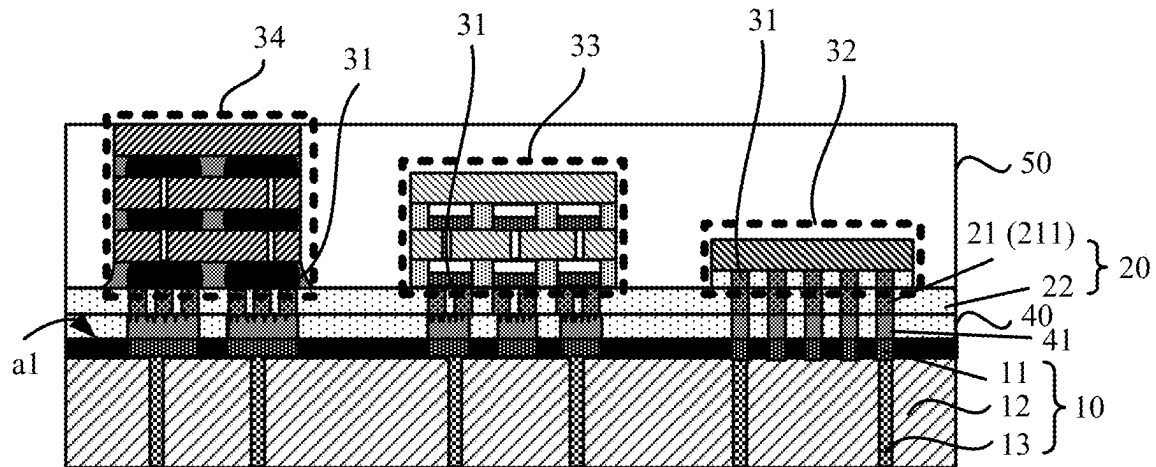
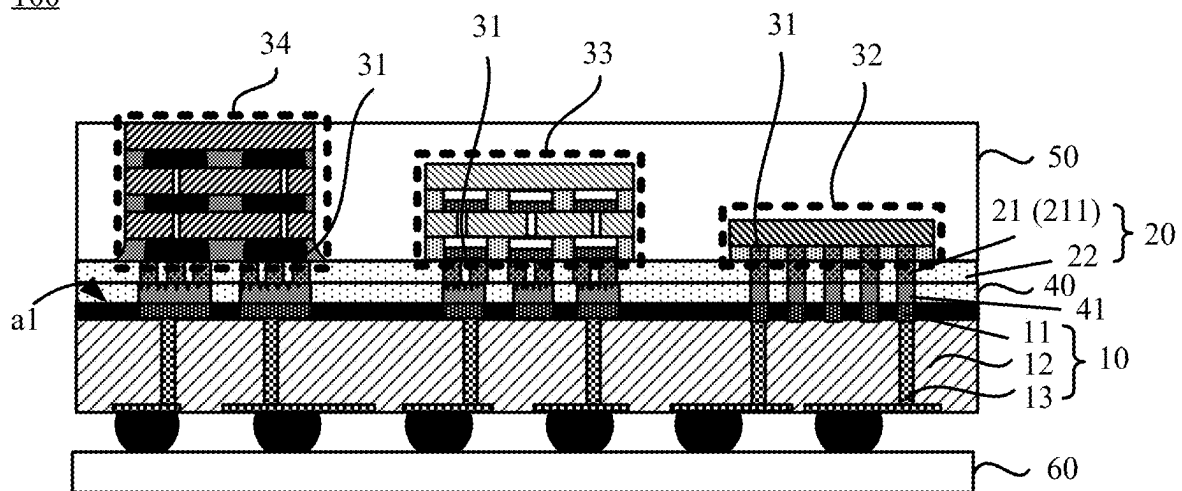


FIG. 7D

100



100



100

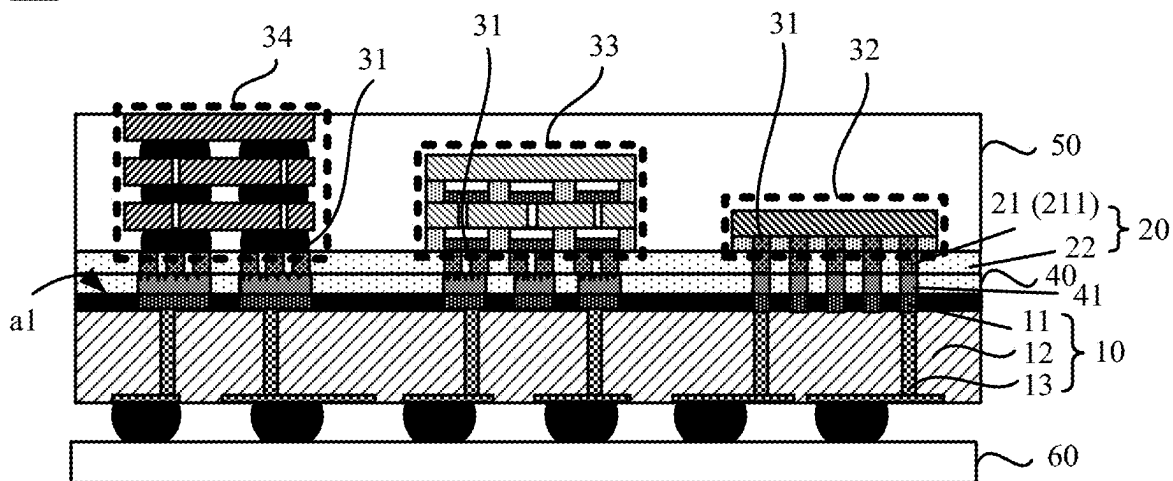


FIG. 9B

100

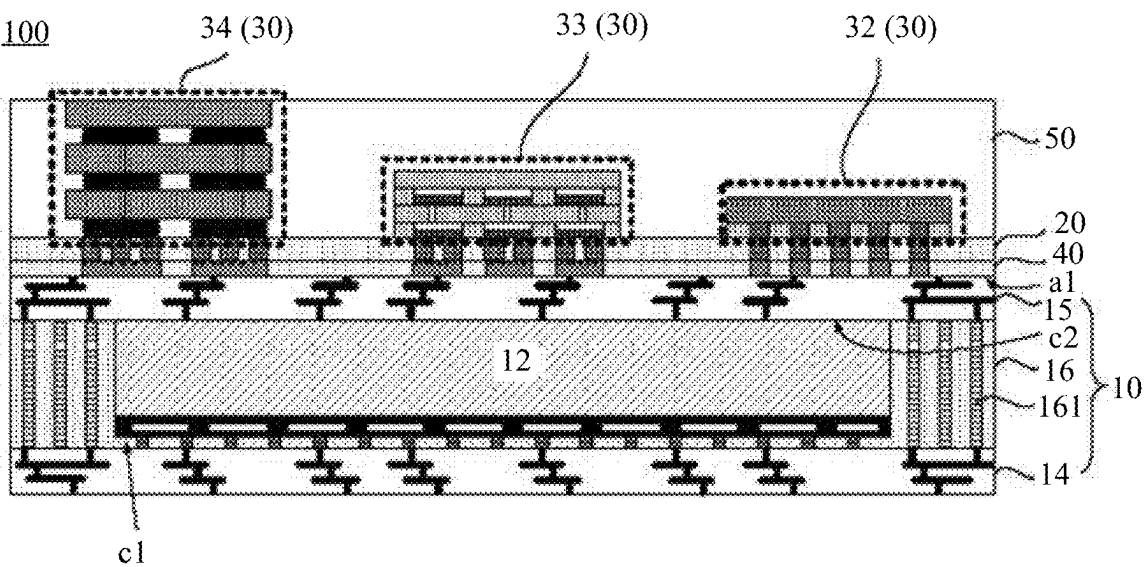


FIG. 10A

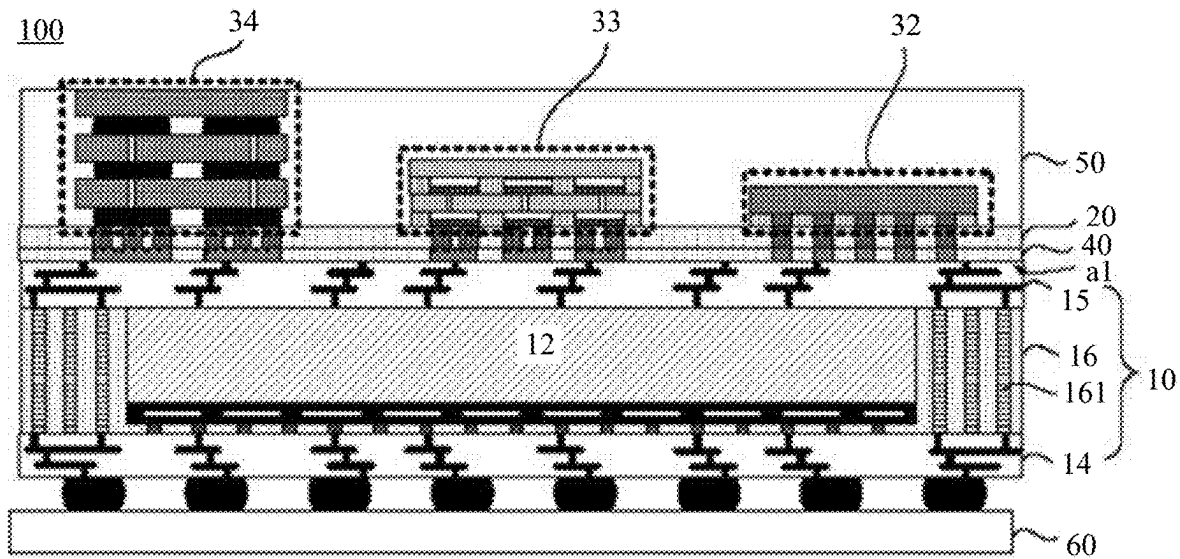


FIG. 10B

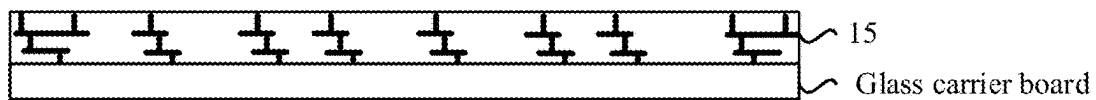


FIG. 11A

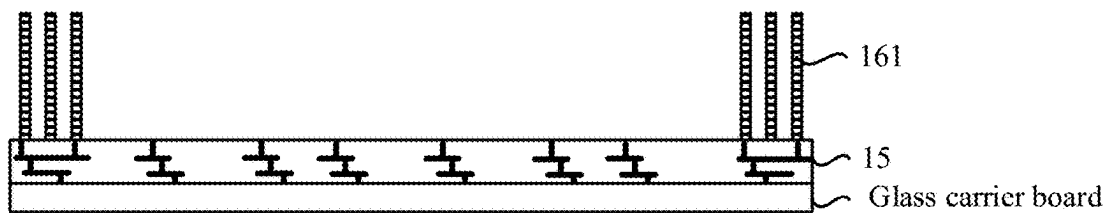


FIG. 11B

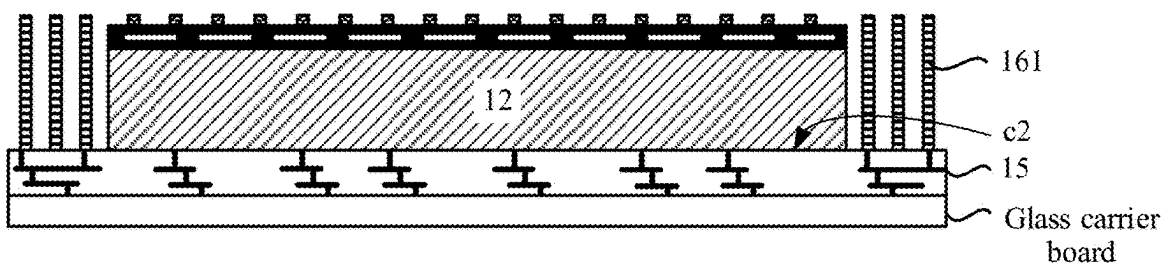


FIG. 11C

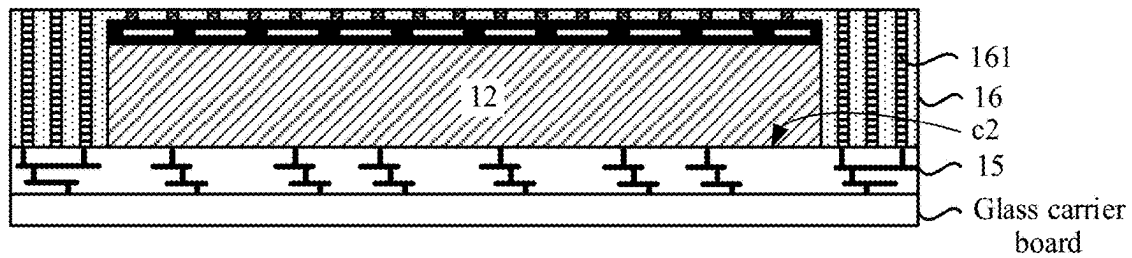


FIG. 11D

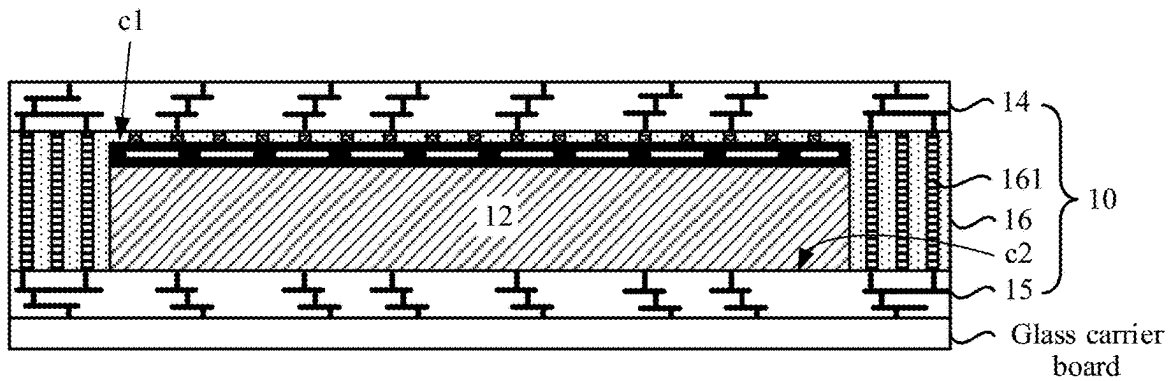


FIG. 11E

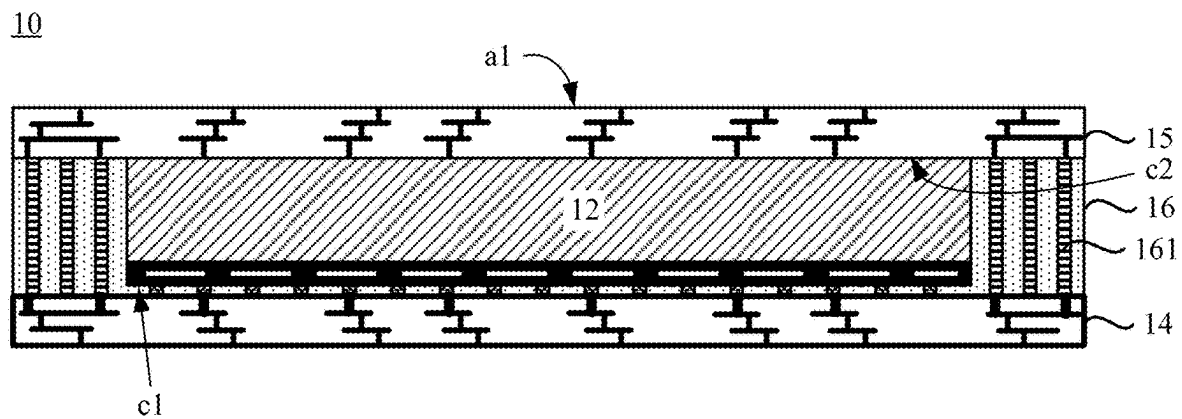


FIG. 11F

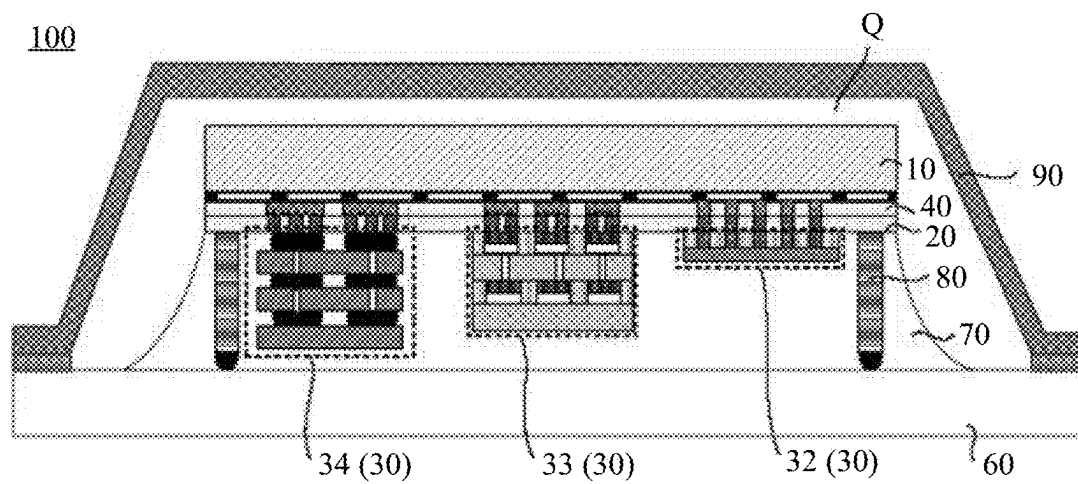


FIG. 12A

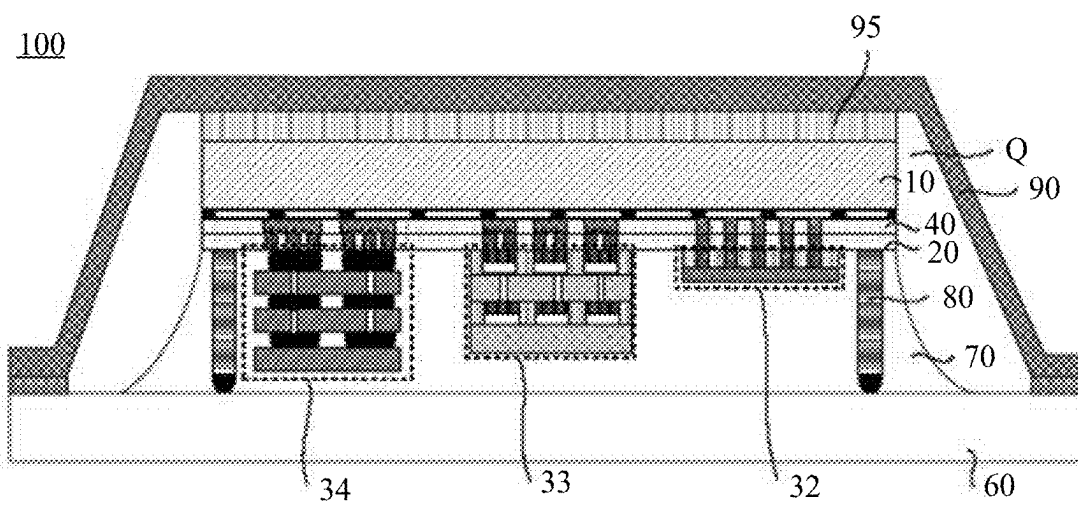


FIG. 12B

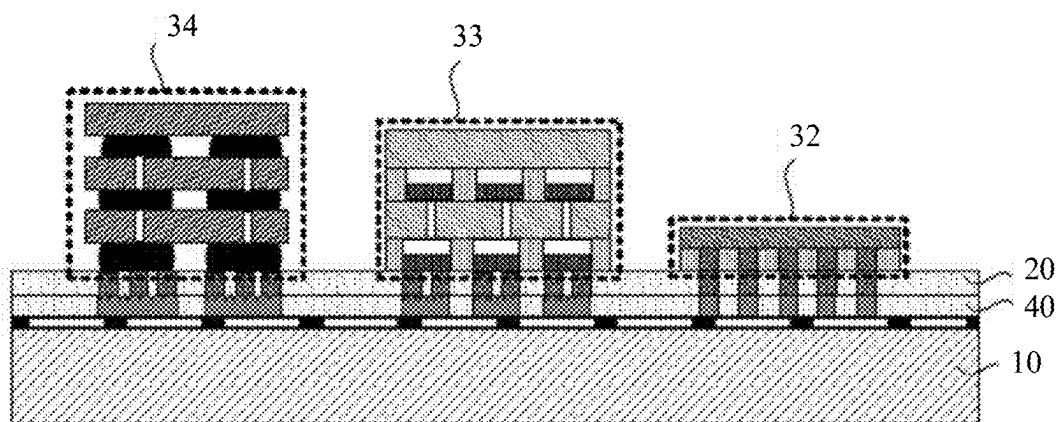


FIG. 13A

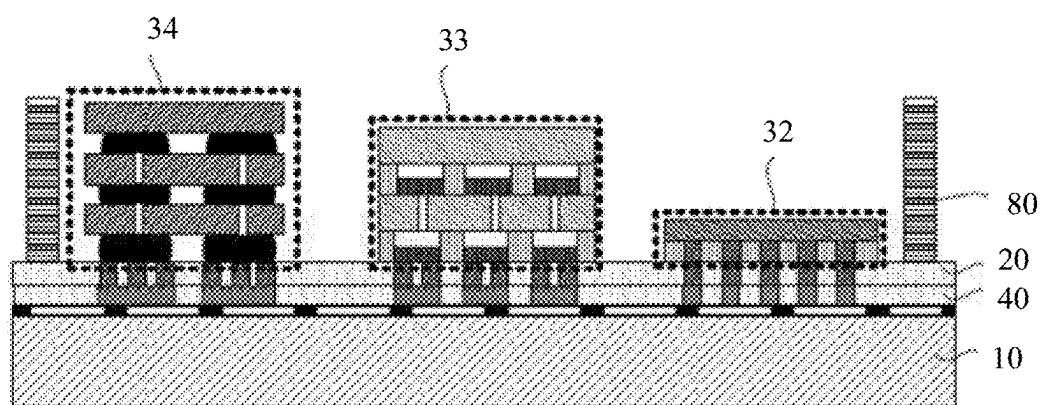


FIG. 13B

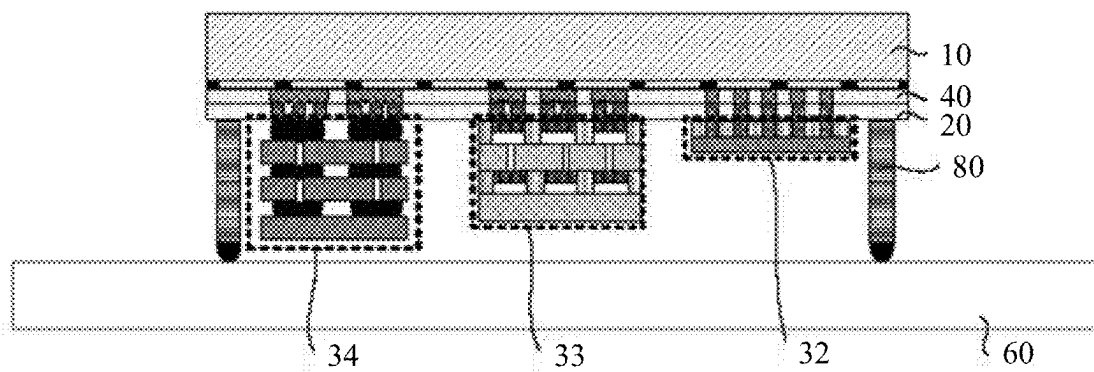


FIG. 13C

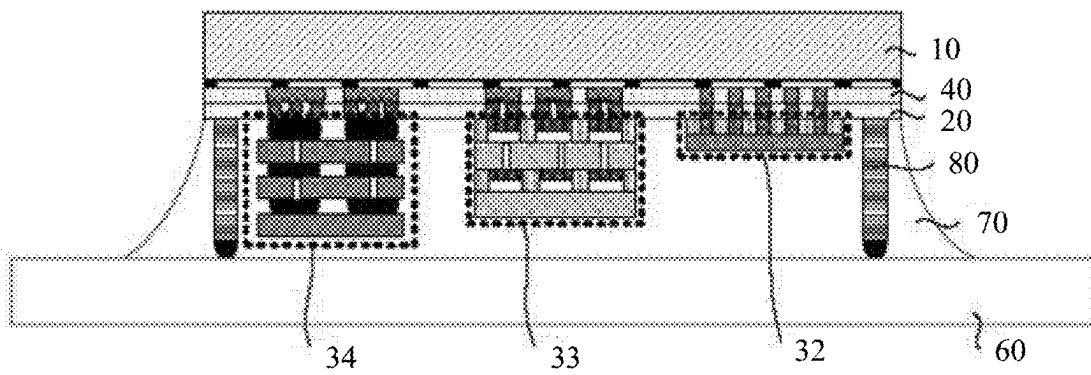


FIG. 13D

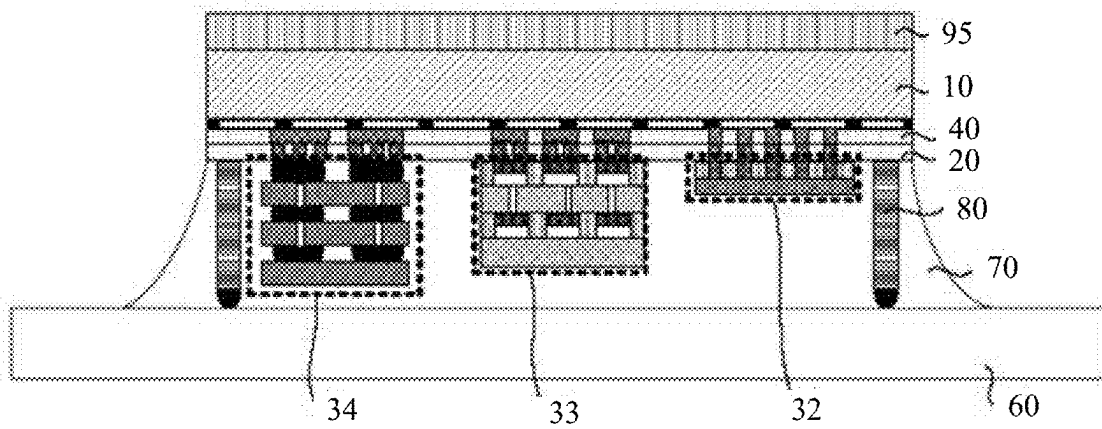


FIG. 13E

100

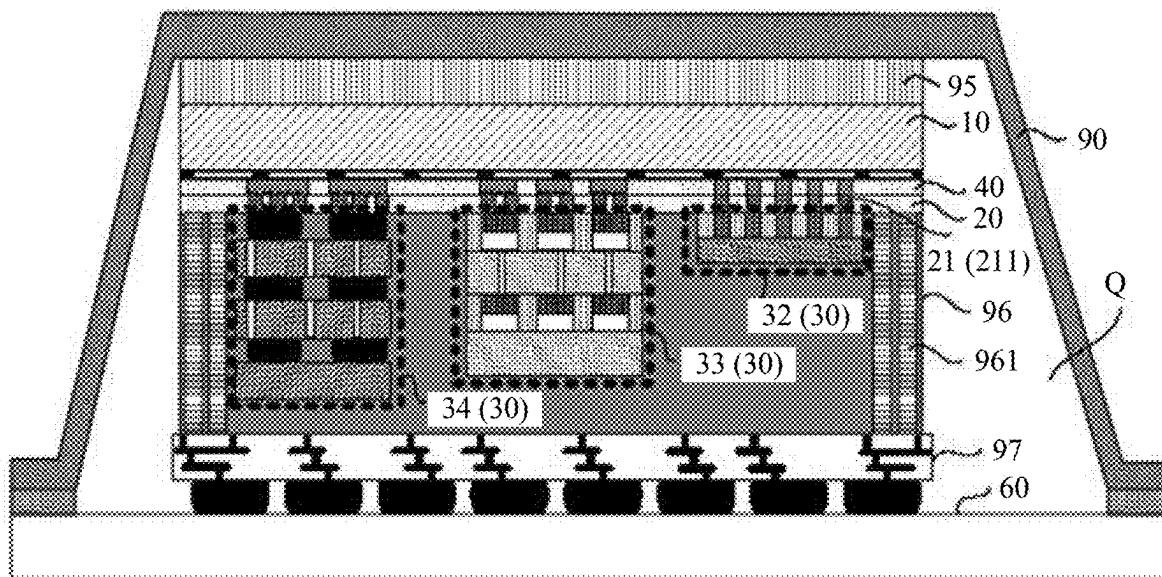


FIG. 14

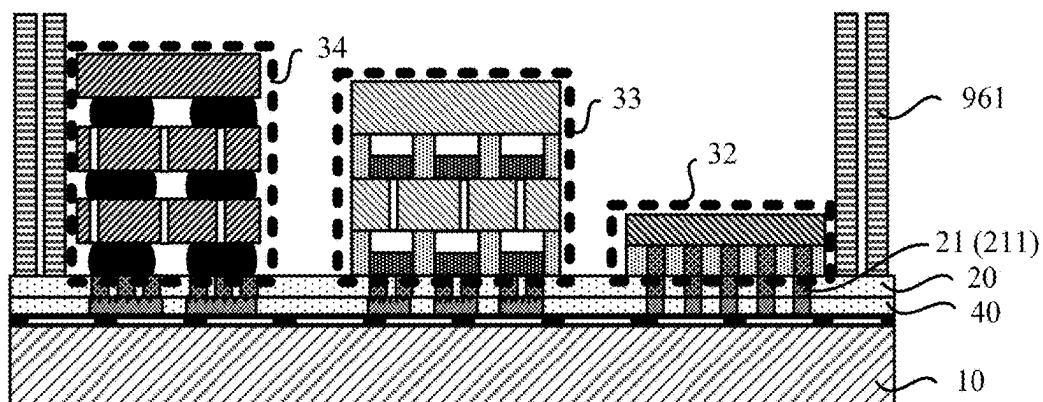


FIG. 15A

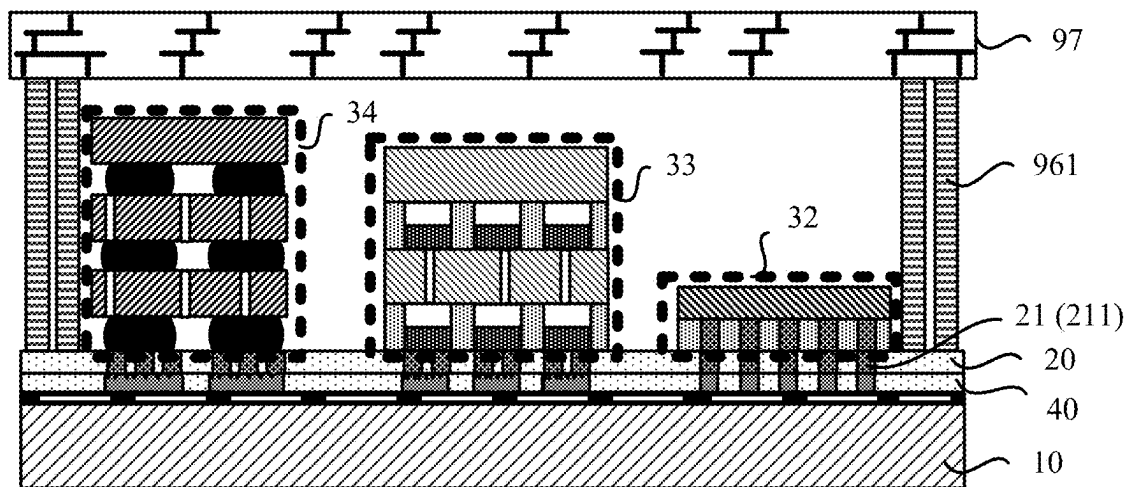


FIG. 15B

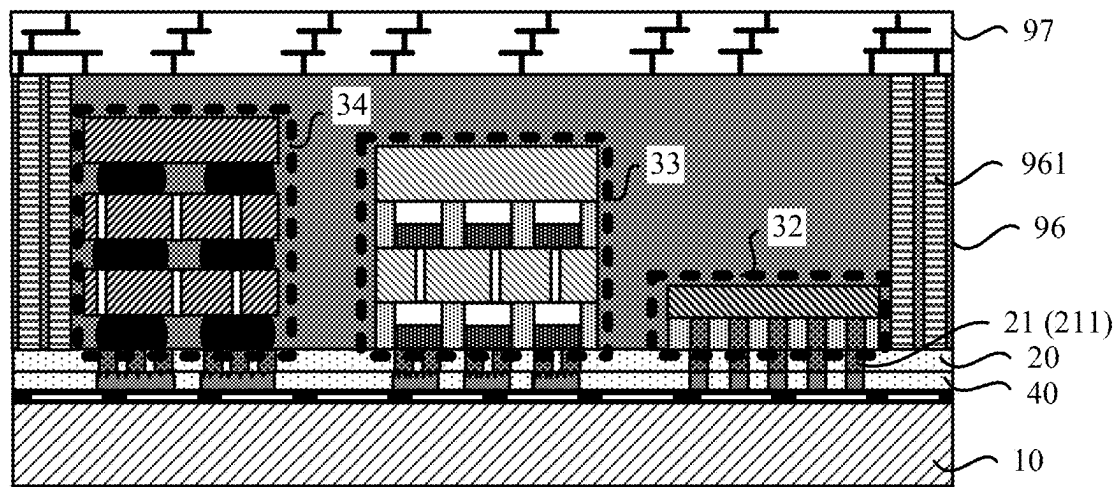


FIG. 15C

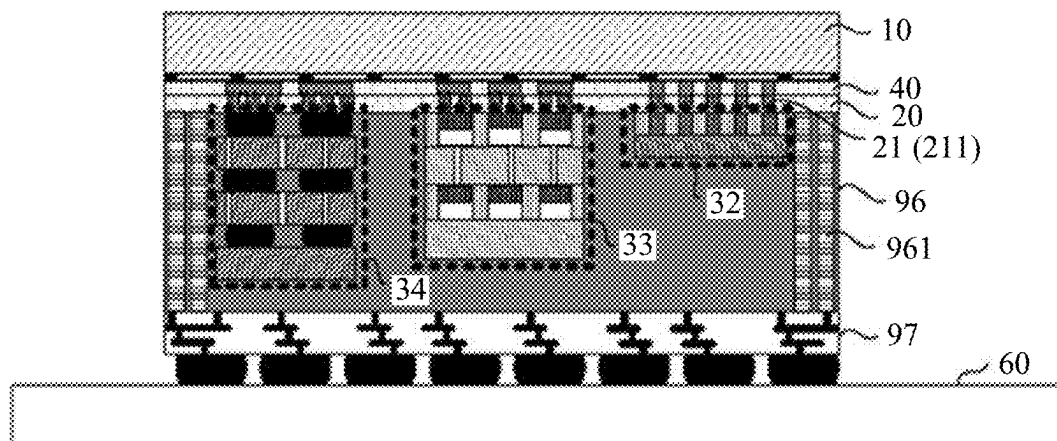


FIG. 15D

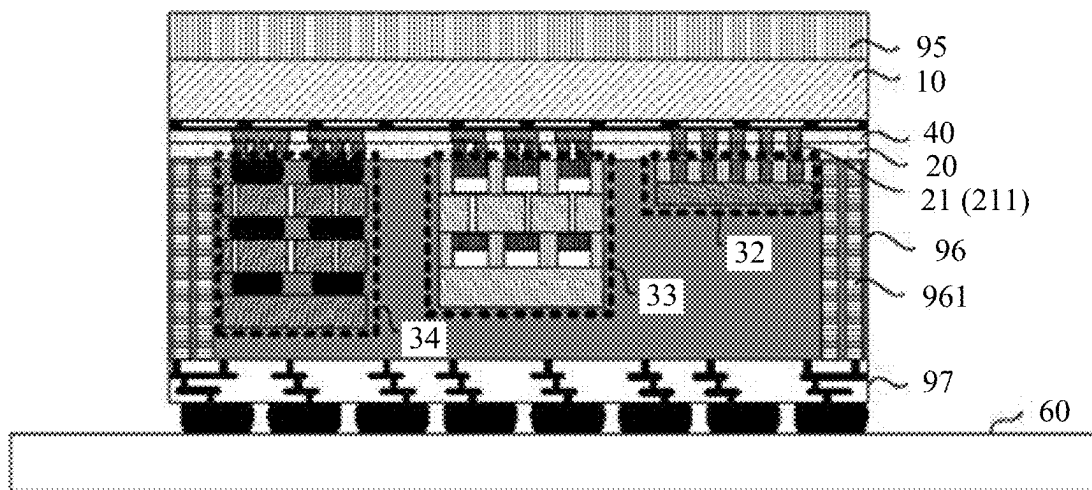


FIG. 15E

1

CHIP PACKAGE ON PACKAGE STRUCTURE, PACKAGING METHOD THEREOF, AND ELECTRONIC DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

This is a continuation of International Patent Application No. PCT/CN2019/114506 filed on Oct. 30, 2019, the disclosure of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

This application relates to the field of electronic technologies, and in particular, to a chip package on package structure, a packaging method thereof, and an electronic device.

BACKGROUND

With development of electronic technologies, electronic devices continuously develop toward miniaturization, integration, and ultra-thinness, and a chip package on package structure in the electronic device also has to develop toward miniaturization.

Package on package (POP) becomes a development trend of the chip package on package structure to allow for miniaturization and integration of the chip package on package structure. Package on package is a structure in which a plurality of secondary chip stack units are bonded to a primary chip stack unit in a same chip package on package structure. However, reliability of bonding the primary chip stack unit and the secondary chip stack unit directly affects performance of the chip package on package structure, and thus becomes a key research topic for persons skilled in the art.

An example in which a plurality of secondary chip stack units whose pin spacings differ greatly are bonded to a primary chip stack unit is used. A chip package on package structure includes four chip stack units, and a first secondary chip stack unit, a second secondary chip stack unit, and a third secondary chip stack unit are bonded to the primary chip stack unit. A pin spacing of the first secondary chip stack unit is less than 1 (micrometer) a pin spacing of the second secondary chip stack unit is 5 and a pin spacing of the third secondary chip stack unit is 40 μm .

Because requirements on a process vary with different pin spacings of chips, different bonding methods are used. A bonding method with a high integration degree needs to be used for a chip having a small pin spacing, and a bonding method with a low integration degree needs to be used for a chip having a large pin spacing. Therefore, how to reliably bond a plurality of secondary chip stack units to a same primary chip stack unit becomes a technical problem to be resolved by the persons skilled in the art.

SUMMARY

Embodiments of this application provide a chip package on package structure, a packaging method thereof, and an electronic device, to resolve how to reliably bond a plurality of secondary chip stack units to a same primary chip stack unit.

To achieve the foregoing objectives, the following technical solutions are used in embodiments.

2

According to a first aspect, a chip package on package structure is provided, including: a primary chip stack unit, having a first surface and a plurality of primary pins insulated and spaced from each other on the first surface; a first bonding layer disposed on the first surface, where the first bonding layer includes a plurality of bonding components insulated and spaced from each other; each of the plurality of bonding components includes at least one bonding part, and any two bonding parts in the plurality of bonding components are insulated and have a same cross-sectional area; and the plurality of bonding components are bonded to the plurality of primary pins respectively, and each of the plurality of primary pins is bonded to at least one bonding part in a bonding component bonded to the primary pin; and a plurality of secondary chip stack units, disposed on a surface that is of the first bonding layer and that is away from the primary chip stack unit, where the secondary chip stack unit has a plurality of micro bumps insulated and spaced from each other, and each of the plurality of micro bumps is bonded to one of the plurality of bonding components, and is bonded to at least one bonding part in a bonding component bonded to the micro bump. In embodiments of this application, the micro bump and the primary pin are bonded by using the bonding component, to separately bond the plurality of secondary chip stack units to the primary chip stack unit. In addition, because any two bonding parts in the first bonding layer have an equal cross-sectional area, when the bonding parts in the first bonding layer are manufactured by using a same process, surfaces of the manufactured bonding parts are planes of similar roughness. In this way, regardless of whether cross-sectional areas of the micro bumps of the plurality of secondary chip stack units are equal, because the surface of the bonding part in contact with the micro bump is a plane, the bonding part may be in close contact with the micro bump. Therefore, the bonding part can be reliably bonded to the micro bump regardless of which bonding process is used. This ensures electrical performance and reliability of the primary chip stack unit and the plurality of secondary chip stack units after bonding.

Optionally, the at least one bonding part is a plurality of bonding parts, and spacings between adjacent bonding parts in a same bonding component in the plurality of bonding components are equal. When the first bonding layer is manufactured by using a chemical-mechanical planarization (CMP) process, a balance between chemical corrosion and mechanical grinding in the CMP process is required, to obtain bonding parts having similar surface flatness. However, when the spacings between adjacent bonding parts are different, surfaces in contact with a polishing liquid have different degrees of chemical corrosion, and the bonding parts obtained through mechanical grinding have different surface flatness. This may affect bonding reliability.

Optionally, cross-sectional areas of the plurality of micro bumps of each secondary chip stack unit in the plurality of secondary chip stack units are not completely equal. The cross-sectional area of the bonding part in the plurality of bonding components is equal to a cross-sectional area of a micro bump having a smallest cross-sectional area. In this case, the bonding part is in a one-to-one correspondence with the micro bump having the smallest cross-sectional area. This ensures reliable bonding of the bonding part and the micro bump. In addition, the bonding part does not need to be excessively small. In this way, a requirement on the manufacturing process of the bonding part can be lowered.

Optionally, a plurality of micro bumps included in each of the plurality of secondary chip stack units have an equal cross-sectional area. This can simplify the process and facilitate manufacturing.

Optionally, a spacing between adjacent micro bumps of a secondary chip stack unit whose micro bump has the smallest cross-sectional area in the plurality of secondary chip stack units is equal to the spacing between adjacent bonding parts in the bonding component. In this way, when a bonding requirement is met, a density of the bonding parts does not need to be large, and the requirement on the manufacturing process is lowered as much as possible. This can reduce manufacturing costs.

Optionally, the chip package on package structure further includes a second bonding layer, and the second bonding layer is disposed between the primary chip stack unit and the first bonding layer. The second bonding layer includes a plurality of auxiliary pins insulated and spaced from each other. Two ends of each of the plurality of auxiliary pins are bonded to each of the plurality of primary pins and each of the plurality of bonding components respectively, and each of the plurality of auxiliary pins is bonded to at least one bonding part in a bonding component bonded to the auxiliary pin. Auxiliary pins are directly bonded to some primary chip stack units sold in the existing markets. This structure may also be applicable to a solution in which the secondary chip stack unit is bonded by using the bonding component in the present disclosure. This can expand an application scope of the chip package on package structure.

Optionally, an auxiliary pin and a micro bump bonded by using a same bonding component in the plurality of bonding components have an equal cross-sectional area. A large cross-sectional area of the auxiliary pin may increase a bonding possibility of adjacent auxiliary pins. A small cross-sectional area of the auxiliary pin may affect a bonding effect of the bonding component and the auxiliary pin, and affect signal transmission on the primary pin.

Optionally, in the plurality of bonding components, a quantity of bonding parts in a bonding component bonded to a micro bump having a large cross-sectional area is greater than a quantity of bonding parts in a bonding component bonded to a micro bump having a small cross-sectional area. In other words, different quantities of bonding parts are disposed based on micro bumps of different sizes, to improve a bonding effect of the bonding component and the micro bump.

Optionally, any two bonding parts in the plurality of bonding components have a same material. When a same manufacturing process is used, the same material of the bonding parts indicates similar surface roughness of manufactured bonding parts. In addition, a plurality of bonding parts can be manufactured by using the same process, and the process is simple and efficient.

Optionally, any two bonding parts in the plurality of bonding components have cross sections of a same shape. The bonding parts have cross sections of the same shape, and the bonding parts during manufacturing have a similar process error. This can improve similarity of surface roughness of the plurality of bonding parts. In addition, this can lower a requirement on a mask plate, simplify the manufacturing process, and reduces costs.

Optionally, bonding components bonded to a same secondary chip stack unit in the plurality of secondary chip stack units include an equal quantity of bonding parts. In this way, whether each bonding component is bonded to a pin

corresponding to the bonding component does not need to be separately considered, which simplifies design and improves efficiency.

Optionally, the plurality of secondary chip stack units include a first secondary chip stack unit, a second secondary chip stack unit, and a third secondary chip stack unit. A micro bump of the first secondary chip stack unit is a copper stud. A micro bump of the second secondary chip stack unit includes a low-temperature solder layer and a high-temperature solder layer bonded to each other, and the low-temperature solder layer is disposed close to the first bonding layer relative to the high-temperature solder layer. A micro bump of the third secondary chip stack unit is a solder ball.

Optionally, the primary chip stack unit includes a primary die, and the primary die has a plurality of first through silicon vias passing through the primary die. An active surface of the primary die is used as the first surface. The plurality of first through silicon vias are bonded to the plurality of primary pins respectively. Because a secondary chip stack unit bonded to the primary chip stack unit is connected to an external signal through the first through silicon via on the primary die, an electrical connection structure does not need to be disposed outside the primary chip stack unit. Therefore, a chip package on package structure of this structure has a small size.

Optionally, the primary chip stack unit includes: a primary die; a first redistribution layer, disposed on the active surface of the primary die and bonded to the primary die; a second redistribution layer, disposed on a non-active surface of the primary die, where a surface that is of the second redistribution layer and that is away from the primary die is used as the first surface; and a first molding layer, disposed between the first redistribution layer and the second redistribution layer and located on a periphery of the primary die, where the first molding layer has a plurality of first through molding vias passing through the first molding layer, and two ends of the plurality of each of first through molding vias are bonded to the first redistribution layer and the second redistribution layer respectively. Because a secondary chip stack unit bonded to the primary chip stack unit is connected to an external signal through the first through molding via disposed on the periphery of the primary die, the primary die does not need to be punched. This can lower the process requirement and reduce costs.

Optionally, the chip package on package structure further includes a second molding layer covering all of the plurality of secondary chip stack units on a side away from the primary chip stack unit. In a direction perpendicular to the primary chip stack unit, a surface of a larger secondary chip stack unit in the plurality of secondary chip stack units is aligned with a surface that is of the second molding layer and that is away from the primary chip stack unit. The second molding layer exposes the surface of the larger secondary chip stack unit in the plurality of secondary chip stack units, so that heat generated by the second molding layer can be dissipated, to improve a heat dissipating effect.

Optionally, the chip package on package structure further includes: a substrate, an underfill layer located between the primary chip stack unit and the substrate and covering the plurality of secondary chip stack units, and a plurality of through dielectric vias embedded in the underfill layer. Two ends of each of the plurality of through dielectric vias are bonded to the substrate and each of the plurality of bonding components respectively. The through dielectric via is directly bonded to the substrate without disposing a redis-

5

tribution layer to lead out signals. This simplifies the process and makes the chip package on package structure light and thin.

Optionally, the chip package on package structure further includes: a third molding layer, covering the plurality of secondary chip stack units, where the third molding layer has a plurality of second through molding vias passing through the third molding layer, and a first end of each of the plurality of second through molding vias is bonded to the bonding part; and a third redistribution layer, disposed on a side that is of the third molding layer and that is away from the primary chip stack unit, where the third redistribution layer is bonded to a second end of each of the plurality of second through molding vias; and a substrate, disposed on a side that is of the third redistribution layer and that is away from the primary chip stack unit, and bonded to the third redistribution layer. The redistribution layer is disposed to lead out electrical signals and is bonded to the substrate. This can increase a quantity of pins and increase a signal transmission amount on the chip package on package structure.

Optionally, the primary chip stack unit includes a primary die, where an active surface of the primary die is used as the first surface. The chip package on package structure further includes: a heat dissipating lid, having a groove, where the heat dissipating lid and the substrate are jointed, the groove and the substrate form a receptacle, and the primary chip stack unit and the plurality of secondary chip stack units are located in the receptacle; and a thermally conductive adhesive, located between the primary chip stack unit and the heat dissipating lid, and configured to bond the primary chip stack unit and the heat dissipating lid. The heat dissipating lid is disposed to improve a heat dissipating effect of the chip package on package structure.

Optionally, each of the plurality of secondary chip stack units includes a plurality of stacked secondary dies, adjacent secondary dies are bonded, and a secondary die closest to the first bonding layer is bonded to the first bonding layer. The secondary chip stack unit may be a chip of a plurality of structures.

According to a second aspect, a packaging method of a chip package on package structure is provided, and includes: A first bonding layer is manufactured in each assembly region of a primary wafer. The primary wafer includes a plurality of assembly regions defined by horizontal and vertical scribe lines. A first surface of the primary wafer has a plurality of primary pins insulated and spaced from each other in each assembly region. The first bonding layer includes a plurality of bonding components insulated and spaced from each other. Each of the plurality of bonding components includes at least one bonding part, and any two bonding parts in the plurality of bonding components are insulated and have a same cross-sectional area. The plurality of bonding components are bonded to the plurality of primary pins respectively, and each of the plurality of primary pins is bonded to at least one bonding part in a bonding component bonded to the primary pin. A plurality of secondary chip stack units are bonded to the first bonding layer. At least two secondary chip stack units are bonded to each assembly region. Each of the plurality of secondary chip stack units has a plurality of micro bumps insulated and spaced from each other. Each of the plurality of micro bumps is bonded to one of the plurality of bonding components, and is bonded to at least one bonding part in a bonding component bonded to the micro bump. Separation is performed along the scribe lines to obtain a plurality of chip package on package structures.

6

Optionally, the manufacturing a first bonding layer in each assembly region of a primary wafer includes: A plurality of bonding studs having an equal cross-sectional area and spaced from each other are formed on the first surface of the primary wafer. Each of the plurality of primary pins is bonded to at least one of the plurality of bonding studs. An insulation film is deposited on the first surface on which the plurality of bonding studs are formed. The insulation film covers the plurality of bonding studs. Chemical-mechanical planarization processing is performed on the insulation film and the plurality of bonding studs, and a plurality of bonding parts and a first insulation layer sleeved on the bonding parts are formed, to obtain the first bonding layer. Bonding parts bonded to a same primary pin are used as one bonding component.

Optionally, the bonding a plurality of secondary chip stack units to the first bonding layer includes: The plurality of secondary chip stack units are sorted in a descending order of bonding temperatures of the micro bumps of the plurality of secondary chip stack units. The secondary chip stack units are sequentially bonded to the first bonding layer.

Optionally, before the manufacturing a first bonding layer in each assembly region of a primary wafer, the packaging method of the chip package on package structure further includes: A second bonding layer is manufactured in each assembly region of the primary wafer. The second bonding layer includes a plurality of auxiliary pins insulated and spaced from each other, and the plurality of auxiliary pins are bonded to the plurality of primary pins respectively.

According to a third aspect, an electronic device is provided, and includes a circuit board and the chip package on package structure according to any one of the first aspect. The chip package on package structure is bonded to the circuit board.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a schematic diagram of a structure of an electronic device according to an embodiment of this application;

FIG. 2A is a top view of a chip package on package structure according to an embodiment of this application;

FIG. 2B is a sectional view in a direction of A1 to A2 in FIG. 2A;

FIG. 2C is a top view of a primary chip stack unit according to an embodiment of this application;

FIG. 2D is a schematic diagram of an arrangement manner of a secondary chip stack unit according to an embodiment of this application;

FIG. 2E is a sectional view in a direction of A1 to A2 in FIG. 2A;

FIG. 3A is a cross-sectional view of a chip package on package structure according to an embodiment of this application;

FIG. 3B is a cross-sectional view of another chip package on package structure according to an embodiment of this application;

FIG. 3C is a top view of a bonding part, a primary pin, and a micro bump according to an embodiment of this application;

FIG. 4 is a schematic top view of a first bonding layer according to an embodiment of this application;

FIG. 5A is a flowchart of a packaging method of a chip package on package structure according to an embodiment of this application;

7

FIG. 5B to FIG. 5F are schematic diagrams of a process of bonding a primary chip stack unit and a secondary chip stack unit according to an embodiment of this application;

FIG. 5G is a flowchart of a method for manufacturing a first bonding layer according to an embodiment of this application;

FIG. 6A is a cross-sectional view of still another chip package on package structure according to an embodiment of this application;

FIG. 6B is a flowchart of another packaging method of a chip package on package structure according to an embodiment of this application;

FIG. 6C is a schematic diagram of surface roughness of an auxiliary solder pad in a second bonding layer according to an embodiment of this application;

FIG. 7A is a cross-sectional view of still another chip package on package structure according to an embodiment of this application;

FIG. 7B is a cross-sectional view of still another chip package on package structure according to an embodiment of this application;

FIG. 7C is a cross-sectional view of still another chip package on package structure according to an embodiment of this application;

FIG. 7D is a cross-sectional view of still another chip package on package structure according to an embodiment of this application;

FIG. 8 is a cross-sectional view of still another chip package on package structure according to an embodiment of this application;

FIG. 9A is a cross-sectional view of still another chip package on package structure according to an embodiment of this application;

FIG. 9B is a cross-sectional view of still another chip package on package structure according to an embodiment of this application;

FIG. 10A is a cross-sectional view of still another chip package on package structure according to an embodiment of this application;

FIG. 10B is a cross-sectional view of still another chip package on package structure according to an embodiment of this application;

FIG. 11A to FIG. 11F are schematic diagrams of a packaging process of a primary chip stack unit according to an embodiment of this application;

FIG. 12A is a cross-sectional view of still another chip package on package structure according to an embodiment of this application;

FIG. 12B is a cross-sectional view of still another chip package on package structure according to an embodiment of this application;

FIG. 13A to FIG. 13E are schematic diagrams of a packaging process of a chip package on package structure according to an embodiment of this application;

FIG. 14 is a cross-sectional view of still another chip package on package structure according to an embodiment of this application; and

FIG. 15A to FIG. 15E are schematic diagrams of a packaging process of a chip package on package structure according to an embodiment of this application.

Reference numerals: 1 electronic device, 2 display module, 3 middle frame, 4 housing, 5 cover, 100 chip package on package structure, 10 primary chip stack unit, 101 primary wafer, 11 primary pin, 12 primary die, 13 first through silicon via, 14 first redistribution layer, 15 second redistribution layer, 16 first molding layer, 161 first through molding via, 20 first bonding layer, 21 bonding component,

8

211 bonding part, 212 bonding stud, 22 first insulation layer, 30 secondary chip stack unit, 31 micro bump, 32 first secondary chip stack unit, 33 second secondary chip stack unit, 34 third secondary chip stack unit, 35 secondary die, 40 second bonding layer, 41 auxiliary solder pad, 50 second molding layer, 60 substrate, 70 underfill layer, 80 through dielectric via, 90 heat dissipating lid, 95 thermally conductive adhesive, 96 third molding layer, 961 second through molding via, and 97 third redistribution layer.

DESCRIPTION OF EMBODIMENTS

Unless otherwise defined, a technical term or a scientific term used in this application should have a general meaning understood by persons skilled in the art. In the specification and claims of this application, the terms “first”, “second”, “third”, and the like are not intended to indicate any order, quantity, or significance, but are intended to distinguish between different components. Therefore, a feature limited by “first”, “second”, or “third” may explicitly or implicitly include one or more features. In the descriptions of embodiments of this application, unless otherwise specified, “a plurality of” means two or more than two.

Orientation terms such as “left”, “right”, “up”, and “down” are defined relative to an orientation at which the components are schematically placed in the accompanying drawings. It should be understood that these orientation terms are relative concepts. These orientation terms are used for corresponding description and clarification, and may vary with the orientation at which the components are placed.

An embodiment of this application provides an electronic device. The electronic device may be a terminal device having a display interface, such as a mobile phone, a television, a display, a tablet, or a vehicle-mounted computer, or may be an intelligent display wearable device such as a smartwatch or a smart band, or may be a communication device such as a server, a memory, or a base station, or may be a smart automobile, or the like. A specific form of the electronic device is not particularly limited in embodiments of this application. For ease of description, an example in which the electronic device is a mobile phone is used for description in the following embodiments.

In this case, as shown in FIG. 1, an electronic device 1 mainly includes a display module 2, a middle frame 3, a housing (or referred to as a battery cover or a rear housing) 4, and a cover 5.

The display module 2 is provided with a light emitting side capable of showing a displayed image and a rear side opposite to the light emitting side. The rear side of the display module 2 is close to the middle frame 3, and the cover 5 is disposed on the light emitting side of the display module 2.

The display module 2 includes a display panel (DP).

In a possible embodiment of this application, the display module 2 is a liquid crystal display module. In this case, the display panel is a liquid-crystal display (LCD). Based on this, the display module 2 further includes a backlight module (BLU) located on a back surface of the liquid crystal display (away from a surface of a side that is of the LCD and that is used to display an image).

The backlight module may provide a light source for the liquid crystal display, so that each sub pixel in the liquid crystal display can emit light to display an image.

Alternatively, in another possible embodiment of this application, the display module 2 is an organic light emitting diode display module. In this case, the display panel is an

organic light-emitting diode (OLED) display. Because an electroluminescence layer is disposed in each sub pixel in the OLED display, the OLED display may implement self-luminescence after receiving a working voltage. In this case, the backlight module does not need to be disposed in the display module 2 having the OLED display.

The cover 5 is located on a side that is of the display module 2 and that is away from the middle frame 3. The cover 5, for example, may be cover glass (CG), and the cover glass may have special toughness.

The middle frame 3 is located between the display module 2 and the housing 4. A surface that is of the middle frame 3 and that is away from the display module 2 is used to install internal elements such as a battery, a printed circuit board (PCB), a camera, and an antenna. After the housing 4 covers the middle frame 3, the internal elements are located between the housing 4 and the middle frame 3.

The electronic device 1 further includes electronic components such as a mainboard, a system on chip (SOC), and a chip package on package structure that are disposed on the PCB. The PCB is configured to carry the electronic components, and complete signal interaction with the electronic components.

To meet a trend of miniaturization, integration, ultra-thinness, and diversified functions of electronic devices, a chip package on package structure gradually develops towards miniaturization and integration.

In view of this, as shown in FIG. 2A (a top view of a chip package on package structure 100), an embodiment of this application provides the chip package on package structure 100, including a primary chip stack unit 10, a first bonding layer 20, and a plurality of secondary chip stack units 30.

As shown in FIG. 2B (a sectional view in a direction of A1 to A2 in FIG. 2A), the primary chip stack unit 10 has a first surface 1a1, and a plurality of primary pins 11 insulated and spaced from each other on the first surface 1a1.

The primary chip stack unit 10 may be an active chip, for example, a logic chip. In this case, the first surface 1a1 of the primary chip stack unit 10 may be an active surface of the primary chip stack unit 10, or a non-active surface of the primary chip stack unit 10. In the primary chip stack unit 10, a side on which a transistor is disposed is referred to as the active surface, and a side without a transistor is referred to as the non-active surface.

Certainly, the primary chip stack unit 10 may also be a passive chip, for example, an interposer.

It should be noted that cross-sectional areas of the plurality of primary pins 11 in the primary chip stack unit 10 may be the same. Certainly, for the cross-sectional areas of the plurality of primary pins 11 in the primary chip stack unit 10, as shown in FIG. 2C (a top view of the primary chip stack unit 10), the cross-sectional areas of the plurality of primary pins 11 in the primary chip stack unit 10 may alternatively be not completely the same.

In some embodiments, as shown in FIG. 2B, cross-sectional areas of primary pins 11 bonded to a same secondary chip stack unit 30 are the same, and cross-sectional areas of primary pins 11 bonded to different secondary chip stack units 30 are different.

In embodiments of this application, a first direction X is a direction from the primary chip stack unit 10 to the secondary chip stack unit 30, or is understood as a direction perpendicular to the primary chip stack unit 10. A second direction Y is perpendicular to the first direction X, or is understood as a direction parallel to the primary chip stack unit 10. A cross-sectional area refers to a cross-sectional area

parallel to the second direction Y, or is understood as an area of an orthographic projection of the primary chip stack unit 10.

A shape of a cross section of the primary pin 11 includes but is not limited to a closed pattern, for example a square, a circle, or a rectangle. In FIG. 2C, the primary pin 11 is shown as a square. In addition, spacings between adjacent primary pins 11 are not limited to the same.

The chip package on package structure includes the plurality of secondary chip stack units 30. FIG. 2B shows an example in which the chip package on package structure includes three secondary chip stack units 30.

The secondary chip stack unit 30 includes a plurality of micro bumps 31 insulated and spaced from each other. A side that is of the secondary chip stack unit 30 and on which the micro bumps 31 are disposed faces the first surface 1a1 of the primary chip stack unit 10.

For the secondary chip stack unit 30 in the chip package on package structure 100, first, an arrangement manner of the plurality of secondary chip stack units 30 is not limited. As shown in FIG. 2A, the plurality of secondary chip stack units 30 may be regularly disposed in parallel. As shown in FIG. 2D (a top view of the chip package on package structure 100), the plurality of secondary chip stack units 30 may be disposed irregularly but properly.

Secondly, sizes of the plurality of secondary chip stack units 30 are not limited. As shown in FIG. 2A, cross-sectional areas of the plurality of secondary chip stack units 30 may be the same. As shown in FIG. 2D, the sizes of the plurality of secondary chip stack units 30 may alternatively be different.

As shown in FIG. 2B, thicknesses h1 of the plurality of secondary chip stack units 30 in the first direction (a direction perpendicular to the primary chip stack unit 10) X may be the same. As shown in FIG. 2E (a sectional view in the direction of A1 to A2 in FIG. 2A), the thicknesses h1 of the plurality of secondary chip stack units 30 in the first direction X may alternatively be different.

Thirdly, as shown in FIG. 2E, quantities of micro bumps 31 included in different secondary chip stack units 30 may be different, and spacings (or referred to as pitch) h2 between adjacent micro bumps 31 may also be different.

Sizes of the plurality of micro bumps 31 included in each secondary chip stack unit 30 may be different. In some embodiments, the sizes of the plurality of micro bumps 31 included in each secondary chip stack unit 30 are the same, to simplify a process and lower a design requirement.

The sizes of the micro bumps 31 of each secondary chip stack unit 30 in the plurality of secondary chip stack units 30 are not completely equal, and the size of the micro bump 31 includes a cross-sectional area of the micro bump 31 and a thickness of the micro bump 31 in the first direction X.

The sizes of the micro bumps 31 of each secondary chip stack unit 30 are not completely equal. In one case, no sizes of the micro bumps 31 of each secondary chip stack unit 30 in the plurality of secondary chip stack units 30 are completely equal. In other words, sizes of micro bumps 31 of any two secondary chip stack units 30 in the secondary chip stack unit 30 are not equal. For example, the chip package on package structure 100 includes four secondary chip stack units 30: a first secondary chip stack unit, a second secondary chip stack unit, a third secondary chip stack unit, and a fourth secondary chip stack unit. Sizes of micro bumps 31 of the four secondary chip stack units are not equal to each other.

The sizes of the micro bumps 31 of each secondary chip stack unit 30 are not completely equal. In another case, the

11

plurality of secondary chip stack units **30** include several secondary chip stack units **30** of which sizes of micro bumps **31** are equal to each other, and further include several secondary chip stack units **30** of which sizes of micro bumps **31** are not equal to each other. For example, the chip package on package structure **100** includes four secondary chip stack units **30**. Sizes of micro bumps **31** of a first secondary chip stack unit and a second secondary chip stack unit are equal, and sizes of micro bumps **31** of the first secondary chip stack unit, a third secondary chip stack unit, and a fourth secondary chip stack unit are not equal to each other.

In addition, materials of micro bump **31** included in different secondary chip stack units **30** may be different.

It should be noted that having size differences within a process error range may be understood as being equal and same in embodiments of this application.

For example, as shown in FIG. 3A (a cross-sectional view of the chip package on package structure **100**), the plurality of secondary chip stack units **30** include a first secondary chip stack unit **32**, a second secondary chip stack unit **33**, and a third secondary chip stack unit **34**.

A micro bump **31** of the first secondary chip stack unit **32** is a copper stud. A periphery of the micro bump **31** is filled with a dielectric material, and the micro bump **31** is embedded in the dielectric material, to implement insulation between adjacent copper studs. Then, the copper stud and the dielectric material are polished by using a CMP process.

Stability of metal and the dielectric material is good, and a spacing between adjacent micro bumps **31** may be, for example, less than 1 μm . The dielectric material may be, for example, silicon dioxide (SiO_2) or silicon nitride (Si_3N_4).

A micro bump **31** of the second secondary chip stack unit **33** is an embedded micro bump. The embedded micro bump includes at least one high-temperature solder layer and at least one low-temperature solder layer (for example, an embedded micro bump includes one high-temperature solder layer and one low-temperature solder layer in FIG. 3A). A solder layer closest to the primary chip stack unit **10** is the low-temperature solder layer. A periphery of the micro bump **31** is filled with a polymer material, and the micro bump **31** is embedded in the polymer material, to implement insulation between adjacent micro bumps **31**. Then, the embedded micro bump and the polymer material are polished by using the chemical-mechanical planarization process.

Herein, the low-temperature solder layer and the high-temperature solder layer are relative, the high-temperature solder layer has a higher melting point than that of the low-temperature solder layer, and the low-temperature solder layer has a lower melting point than that of the high-temperature solder layer.

A spacing between adjacent micro bumps **31** may be, for example, about 5 μm . A material constituting the high-temperature solder layer may be, for example, copper (Cu), and a material constituting the low-temperature solder layer may be, for example, tin (Sn). The polymer material may be, for example, benzocyclobutene (BCB) or polyimide (PI).

A micro bump **31** of the third secondary chip stack unit **34** is a solder ball. A periphery of the solder ball is filled with a filling adhesive, to implement insulation between adjacent solder balls. A spacing between adjacent micro bumps **31** may be, for example, about 40 μm . The solder ball may be made of a material, for example, Cu/Sn or Cu/Sn/silver (Ag).

The micro bump **31** of the first secondary chip stack unit **32** has a smallest cross-sectional area, and the micro bump **31** of the third secondary chip stack unit **34** is a largest cross-sectional area.

12

The first secondary chip stack unit **32**, the second secondary chip stack unit **33**, and the third secondary chip stack unit **34** may be devices such as a memory (e.g., double data rate (DDR)), an intellectual property (IP) core, or a silicon chip (e.g., a silicon die). This can greatly shorten a process period, reduce costs, and effectively implement heterogeneous integration.

The first secondary chip stack unit **32**, the second secondary chip stack unit **33**, and the third secondary chip stack unit **34** may be bonded to the primary chip stack unit **10** after a good product test, to ensure a product yield. A chip that passes the good product test is referred to as a known good die (KGD).

As shown in FIG. 2B, the first bonding layer **20** is located between the primary chip stack unit **10** and the secondary chip stack unit **30**.

In other words, the first bonding layer **20** is disposed on the first surface of the primary chip stack unit **10**, and the secondary chip stack unit **30** is disposed on a surface that is of the first bonding layer **20** and that is away from the primary chip stack unit **10**.

The first bonding layer **20** includes a plurality of bonding components **21** insulated and spaced from each other. Each bonding component **21** in the plurality of bonding components **21** includes at least one bonding part **211**, and any two bonding parts **211** in the plurality of bonding components **21** are insulated and spaced from each other.

For example, a periphery of the bonding part **211** may be filled with an insulation material, to form a first insulation layer **22** sleeved on the bonding part **211**. In this case, any two bonding parts **211** are insulated, and the bonding part **211** passes through the first insulation layer **22** in the first direction X. The insulation material may be, for example, the polymer material or the dielectric material.

As shown in FIG. 2B, the plurality of bonding components **21** are separately bonded to the plurality of primary pins **11** and the plurality of micro bumps **31**. Each primary pin **11** in the plurality of primary pins **11** is bonded to at least one bonding part **211** in a bonding component **21** bonded to the primary pin **11**. Each micro bump **31** in the plurality of micro bumps **31** is bonded to at least one bonding part **211** in a bonding component **21** bonded to the micro bump **31**.

Herein, the plurality of bonding components **21** are separately bonded to the plurality of primary pins **11** and the plurality of micro bumps **31**. It may alternatively be understood that, as shown in FIG. 2B, two ends of each bonding component **21** are bonded to one primary pin **11** and one micro bump **31** respectively, to bond the primary pin **11** and the micro bump **31**. It may be understood that, to ensure bonding of the primary pin **11** and the micro bump **31**, a primary pin **11** and a micro bump **31** that are bonded to a same bonding component **21** should be bonded to at least one same bonding part **211** in the bonding component **21**.

Bonding refers to a process in which two homogeneous or heterogeneous materials are directly combined under a specific condition after surface processing, to electrically or mechanically interconnect the two materials.

A manner of bonding the bonding component **21** and the primary pin **11** may be, for example, direct contact bonding when the bonding component **21** is manufactured. A manner of bonding the micro bump **31** and the bonding component **21** may be processes such as micro bump bonding, embedded micro bump bonding, hybrid bonding (HB), surface activated bonding (SAB), atomic diffusion bonding (ADB), and wire bonding (WB), to implement bonding.

For a secondary chip stack unit **30** that has a high bandwidth/bit width requirement, processes such as embed-

13

ded micro bump bonding, hybrid bonding, surface activated bonding, or atomic diffusion bonding may be used, to reduce a spacing between bump micros **31**, improve an integration degree, and reduce a delay. For a secondary chip stack unit **30** that does not have a high bandwidth/bit width requirement, processes such as wire bonding or micro bump bonding may be used, to reduce costs.

The micro bump **31** of the first secondary chip stack unit **32** in FIG. 3A is the copper stud. The first secondary chip stack unit **32** uses a hybrid bonding process to bond the micro bump **31** and the bonding part **211** by using thermo compression bonding. The hybrid bonding has a very high requirement on cleanliness and roughness of a bonding interface, and the roughness is required as the order of 0.5 nm.

When a material of the bonding part **211** is copper, a material of the micro bump **31** is the same as the material of the bonding part **211**, and bonding of the micro bump **31** and the bonding part **211** is metal-to-metal direct bonding. When the insulation material filled on the periphery of the bonding part **211** is a dielectric material, the dielectric material on the periphery of the micro bump **31** and the dielectric material on the periphery of the bonding part **211** may be further bonded.

The micro bump **31** of the second secondary chip stack unit **33** is the embedded micro bump. The second secondary chip stack unit **33** uses an embedded micro bump bonding process to bond the micro bump **31** and the bonding part **211** by using thermo compression or high temperature reflow. A material of the micro bump **31** is different from the material of the bonding part **211**, and bonding of the micro bump **31** and the bonding part **211** is eutectic bonding.

When the insulation material filled on the periphery of the bonding part **211** is a polymer material, the polymer material on the periphery of the micro bump **31** and the polymer material on the periphery of the bonding part **211** may be further bonded.

The micro bump **31** of the third secondary chip stack unit **34** is the solder ball. The third secondary chip stack unit **34** uses a micro bump bonding process to bond the micro bump **31** and the bonding part **211** by using thermo compression or high temperature reflow.

A material of the micro bump **31** is different from the material of the bonding part **211**, and bonding of the micro bump **31** and the bonding part **211** is eutectic bonding.

For a quantity of the bonding parts **211** included in the bonding component **21**, in some embodiments, quantities of bonding parts **211** included in all bonding components **21** are equal as shown in FIG. 3B (a cross-sectional view of the chip package on package structure **100**). FIG. 3B shows an example in which the bonding component **21** includes one bonding part **211**.

In other embodiments, quantities of bonding parts **211** included in all bonding components **21** are not completely equal as shown in FIG. 3C (a top view of the chip package on package structure **100**).

For example, in the plurality of bonding components **21**, a quantity of bonding parts **211** in a bonding component **21** bonded to a micro bump **31** having a large cross-sectional area is greater than a quantity of bonding parts **211** in a bonding component **21** bonded to a micro bump **31** having a small cross-sectional area as shown in FIG. 3C.

In other words, a micro bump **31** has a larger cross-sectional area, indicating that a bonding component **21** bonded to the micro bump **31** includes a larger quantity of bonding parts **211**. For example, a micro bump **31** of the third secondary chip stack unit **34** has a largest cross-

14

sectional area, and a bonding component **21** bonded to the micro bump **31** includes three bonding parts **211**. A micro bump **31** of the second secondary chip stack unit **33** has a second largest cross-sectional area, and a bonding component **21** bonded to the micro bump **31** includes two bonding parts **211**. A micro bump **31** of the first secondary chip stack unit **32** has a smallest cross-sectional area, and a bonding component **21** bonded to the micro bump **31** includes one bonding part **211**.

In this way, the quantity of bonding parts **211** in the bonding component **21** may be adjusted based on the cross-sectional area of the micro bump **31**, and a contact area between each bonding component **21** and each micro bump **31** is not very small. This ensures a bonding effect between the bonding component **21** and the micro bump **31**.

In some embodiments, bonding components **21** bonded to a same secondary chip stack unit **30** include an equal quantity of bonding parts **211** as shown in FIG. 3C, to reduce types of bonding components **21** and reduce process difficulty.

As shown in FIG. 4 (a top view of the first bonding layer **20**), quantities of bonding parts **211** included in different bonding components **21** may not be the same, but any two bonding parts **211** in the first bonding layer **20** have a same cross-sectional area.

For example, in some embodiments, any two bonding parts **211** in the first bonding layer **20** have cross sections of a same shape as shown in FIG. 4. For example, a shape of a cross section of the bonding part **211** is a closed pattern, for example, a circle or a rectangle.

In embodiments of this application, a cross-sectional area and a shape of a cross section of the bonding part **211** are not limited.

In a fixed range, if the bonding part **211** has a large cross-sectional area, adjacent bonding parts **211** may have a small spacing and may be bonded. In some embodiments, a cross-sectional area of the bonding part **211** is less than or equal to a cross-sectional area of a micro bump **31** having a smallest cross-sectional area, to reduce a bonding probability of adjacent bonding parts **211**.

For example, when cross-sectional areas of micro bumps **31** in the plurality of secondary chip stack units are different, a cross-sectional area of the bonding part **211** is equal to a cross-sectional area of a micro bump **31** having a smallest cross-sectional area.

Because the bonding part **211** has a large cross-sectional area, adjacent bonding parts **211** may have a small spacing and may be bonded. A small cross-sectional area of the bonding part **211** may increase difficulty of a manufacturing process. Therefore, in embodiments of this application, the cross-sectional area of the bonding part **211** is equal to the cross-sectional area of the micro bump **31** having the smallest cross-sectional area. This can implement a largest bonding contact surface between the bonding part **211** and the micro bump **31** having the smallest cross-sectional area, to improve a bonding effect of the two. In addition, the cross-sectional area of the bonding part **211** does not need to be excessively large or small, to reduce the bonding probability of the adjacent bonding parts **211** and reduce process difficulty.

As shown in FIG. 3A, the micro bump **31** (e.g., a copper stud) of the first secondary chip stack unit **32** has the smallest cross-sectional area, and the cross-sectional area of the bonding part **211** is equal to a cross-sectional area of the micro bump **31** of the first secondary chip stack unit **32**.

In the chip package on package structure **100** shown in FIG. 3A, bonding of the plurality of secondary chip stack

15

units **30** and primary chip stack unit **10** is chip-to-chip bonding (chip-to-chip). As shown in FIG. 5A, the following is a packaging method of the chip package on package structure **100** formed through bonding the plurality of secondary chip stack units **30** and the primary chip stack unit **10**.

S10: As shown in FIG. 5B, manufacture a first bonding layer **20** in each assembly region S of a primary wafer **101**, where the primary wafer **101** includes a plurality of assembly regions S defined by horizontal and vertical scribe lines.

The first bonding layers **20** in the plurality of assembly regions S may be obtained through manufacturing by using a same process. An assembly region S is used as an example.

From a cross-sectional view as shown in FIG. 5C, a bonding part **21** in the first bonding layer **20** is bonded to a primary pin **11** in the primary wafer **101**.

S20: Bond a micro bump **31** of a secondary chip stack unit **30** and the bonding component **21**, to separately bond at least two secondary chip stack units **30** in each assembly region S to the first bonding layer **20**. FIG. 3A shows one assembly region S as an example.

A same type of secondary chip stack units **30** (for example, a first secondary chip stack unit **32** in each assembly region S) in different assembly regions S may be simultaneously bonded in a single bonding process.

In this case, bonding of the secondary chip stack unit **30** and the primary wafer **101** is chip-to-wafer bonding (chip-to-wafer). Wafer refers to a silicon chip used for manufacturing a silicon semiconductor integrated circuit. The silicon chip is circular, and therefore it is referred to as a wafer.

S30: Perform separation along the scribe lines to obtain a plurality of chip package on package structures **100**.

In this case, bonding of the obtained chip package on package structures **100** is chip-to-chip bonding.

An assembly region S is used as an example. In some embodiments, as shown in FIG. 5G, step S10 includes the following steps:

S11: As shown in FIG. 5D, form a plurality of bonding studs **212** having an equal cross-sectional area and spaced from each other on a first surface of the primary wafer **101**. Each primary pin **11** is bonded to at least one bonding stud **212**.

For example, the bonding stud **212** may be formed by using an electroplating process. That the bonding stud **212** is formed by using an electroplating process includes:

First, a seed layer is sputtered on the first surface of the primary wafer **101**. A material of the seed layer may be, for example, titanium (Ti) or Cu. A thickness of the seed layer may be, for example, 100 nm to 200 nm.

Then, as shown in FIG. 5E, a photoresist is spin-coated on the seed layer, and an opening is made through exposure and development at a position of a bonding stud **212** to be formed. A size of the opening matches a size of the bonding stud **212** to be formed.

Then, a structure in FIG. 5E is placed in an electroplating solution for electroplating, and the electroplating is stopped when the electroplated structure reaches a proper thickness. The photoresist is removed, and a seed layer not covered by the bonding stud **212** is removed, to form the bonding stud **212** shown in FIG. 5D. The bonding stud **212** includes a part electroplated in the electroplating solution and a seed layer part.

S12: As shown in FIG. 5F, form an insulation film covering the bonding stud **212** on the first surface of the primary wafer **101** on which the bonding stud **212** is formed.

For example, the insulation film may be formed by using a chemical vapor deposition (CVD) method.

16

S13: Perform chemical-mechanical planarization processing on the insulation film and the bonding stud **212**, to obtain the first bonding layer **20**.

For example, the insulation film and the bonding stud **212** are polished until a bonding part **211** with a proper thickness is left, and a plurality of bonding parts **211** and a first insulation layer **22** sleeved on the bonding parts **211** shown in FIG. 5C are formed, to obtain the first bonding layer **20**. Bonding parts **211** bonded to a same primary pin **11** are used as a bonding component **21**.

During manufacturing of the first bonding layer **20**, to improve uniformity of surface roughness of each bonding part **211** in the first bonding layer **20**, in some embodiments, when a bonding component **21** includes a plurality of bonding parts **211**, spacings **h3** between adjacent bonding parts **211** in a same bonding component **21** are equal as shown in FIG. 4.

When the first bonding layer **20** is manufactured by using the CMP process, a balance between chemical corrosion and mechanical grinding in the CMP process is required, to obtain bonding parts **211** having similar surface flatness. However, when the spacings **h3** between adjacent bonding parts **211** are different, surfaces in contact with a polishing liquid have different degrees of chemical corrosion, and the bonding parts **211** obtained through mechanical grinding have different surface flatness. Therefore, the spacings **h3** between adjacent bonding parts **211** in the same bonding component **21** are set to be equal. This can improve uniformity of surface roughness of all bonding parts **211** in the first bonding layer **20**, to improve reliability of bonding the bonding part **211** and the micro bump **31**.

In some embodiments, a spacing **h2** between adjacent micro bumps **31** of a secondary chip stack unit **10** having micro bumps **31** of a smallest cross-sectional area is equal to a spacing **h3** between adjacent bonding parts **211** in a same bonding component **21**.

The chip package on package structure **100** shown in FIG. 3A is used as an example. The micro bump **31** (e.g., a copper stud) of the first secondary chip stack unit **32** has a smallest cross-sectional area. In this case, the spacing **h3** between adjacent bonding parts **211** in the same bonding component **21** is equal to a spacing **h2** between micro bumps **31** of the first secondary chip stack unit **32**.

In this case, the bonding part **211** is in a one-to-one correspondence with the micro bump **31** having the smallest cross-sectional area. This ensures reliable bonding of the bonding part **211** and the micro bump **31**. In addition, the bonding part **211** and the spacing **h2** between the bonding parts **211** do not need to be excessively small. In this way, requirements on the manufacturing process of the bonding part **211** can be lowered.

In some chip package on package structures **100**, bonding temperatures are different when micro bumps **31** of a plurality of secondary chip stack units **30** are bonded to the first bonding layer **20**. Therefore, in some embodiments, step S20 includes: sort the plurality of secondary chip stack units **30** in a descending order of bonding temperatures of the micro bumps **31** of the plurality of secondary chip stack units **30**, and sequentially bond the plurality of secondary chip stack units **30** to the first bonding layer **20**.

The chip package on package structure **100** shown in FIG. 3A is used as an example. The first secondary chip stack unit **32** is bonded to the first bonding layer **20** by using a hybrid bonding process, and a bonding temperature is 220° C. to 250° C. The second secondary chip stack unit **33** is bonded to the first bonding layer **20** by using an embedded micro bump bonding process, and a bonding temperature is 220°

17

C. or below. The third secondary chip stack unit **34** is bonded to the first bonding layer **20** by using a micro bump bonding process, and a bonding temperature is 260° C. or above.

The first secondary chip stack unit **32**, the second secondary chip stack unit **33**, and the third secondary chip stack unit **34** are sorted in the descending order of the bonding temperatures of the micro bumps **31**, and a sorting order is: the third secondary chip stack unit **34**, the first secondary chip stack unit **32**, and the second secondary chip stack unit **33**.

When the three secondary chip stack units are separately bonded to the first bonding layer **20**, the third secondary chip stack unit **34** in the assembly region S is first bonded to the first bonding layer **20** in a same bonding process. Then, the first secondary chip stack unit **32** in each assembly region S is bonded to the first bonding layer **20** in a same bonding process. Finally, the second secondary chip stack unit **33** in each assembly region S is bonded to the first bonding layer **20** in a same bonding process.

In embodiments of this application, the micro bump **31** and the primary pin **11** are bonded by using the bonding component **21**, to separately bond the plurality of secondary chip stack units **30** to the primary chip stack unit **10**. In addition, because any two bonding parts **211** in the first bonding layer **20** have an equal cross-sectional area (or understood as having a same feature size), when the bonding parts **211** in the first bonding layer **20** are manufactured by using a same process, surfaces of the manufactured bonding parts **211** are planes of similar roughness. In this way, regardless of whether cross-sectional areas of the micro bumps **31** of the plurality of secondary chip stack units **30** are equal, because the surface of the bonding part **211** in contact with the micro bump **31** is a plane, the bonding part **211** may be in close contact with the micro bump **31**. Therefore, regardless of which bonding process is used, whether the cross-sectional areas of the micro bumps **31** are the same, or whether the spacings between adjacent micro bumps **31** are equal, the bonding part **211** can be reliably bonded to the micro bump **31**. This ensures bonding strength of the primary chip stack unit **10** and the plurality of secondary chip stack units **30**, and electrical performance and reliability after bonding.

Based on any one of the foregoing chip package on package structures **100**, as shown in FIG. 6A (a cross-sectional view of the chip package on package structure **100**), the chip package on package structure **100** further includes a second bonding layer **40**, and the second bonding layer **40** is disposed between the primary chip stack unit **10** and the first bonding layer **20**.

The second bonding layer **40** includes a plurality of auxiliary pins **41** insulated and spaced from each other. Two ends of each of the plurality of auxiliary pins **41** are bonded to each of the plurality of primary pins **11** and each of the plurality of bonding components **21** respectively, and each of the plurality of auxiliary pins **41** is bonded to at least one bonding part **211** in a bonding component **21** bonded to the auxiliary pin **41**.

To form the chip package on package structure **100** shown in FIG. 6A, as shown in FIG. 6B, before the first bonding layer **20** is formed, the packaging method of the chip package on package structure further includes the following step:

S05: Manufacture the second bonding layer **40** in each assembly region S of the primary wafer **101**.

In this case, step **S10** includes: manufacturing the first bonding layer **20** on the second bonding layer **40** in each assembly region S.

18

A process for manufacturing the second bonding layer **40** may be the same as the process for manufacturing the first bonding layer **20**. As shown in FIG. 6A, the second bonding layer **40** finally formed includes a second insulation layer **42** sleeved on the auxiliary pin **41**. In this case, any two auxiliary pins **41** are insulated, and the auxiliary pin **41** passes through the second insulation layer **42** in the first direction X.

The second bonding layer **40** is directly formed on some primary chip stack units sold in the existing markets. This structure may also be applicable to a solution in which the secondary chip stack unit **30** is bonded by using the bonding component **21** in the present disclosure. This can expand an application scope of the chip package on package structure **100**.

A large cross-sectional area of the auxiliary pin **41** may increase a bonding possibility of adjacent auxiliary pins **41**. A small cross-sectional area of the auxiliary pin **41** may affect a bonding effect of the bonding component **21** and the auxiliary pin **41**, and affect signal transmission on the primary pin **31**. Therefore, in some embodiments, the auxiliary pin **41** and the micro bump **31** bonded by using the bonding component **21** have an equal cross-sectional area as shown in FIG. 6A.

It should be noted that, as shown in FIG. 6C, when the second bonding layer **40** is manufactured by using the process of manufacturing the first bonding layer **20**, because cross-sectional areas of all auxiliary pins **41** differ greatly, excessive chemical-mechanical planarization processing is performed on a surface of an auxiliary pin **41** having a largest cross-sectional area when a surface of an auxiliary pin **41** having a smallest cross-sectional area meets a roughness requirement. Therefore, surfaces of the auxiliary pins **41** manufactured by using the CMP process differ greatly in roughness. In this case, when a bonding stud **212** is manufactured by using an electroplating process, a thickness of a formed seed layer may be adjusted based on the roughness of the auxiliary pin **41**, to ensure obtaining a bonding stud **212** having an ideal thickness.

A structure of the chip package on package structure **100** provided in embodiments of this application is described below by using several detailed embodiments as examples.

Embodiment 1

As shown in FIG. 7A (a cross-sectional view of a chip package on package structure **100**), the chip package on package structure **100** includes a primary chip stack unit **10**, a first bonding layer **20**, a plurality of secondary chip stack units **30**, a second bonding layer **40**, and a second molding layer **50**.

The primary chip stack unit **10** includes a primary die (e.g., a bare die) **12**. The primary die **12** has a plurality of first through silicon vias (TSVs) **13** passing through the primary die **12**. An active surface of the primary die **12** is used as a first surface **al**. The first through silicon via **13** is bonded to a primary pin **11**.

The second bonding layer **40** is located on the first surface **al** of the primary chip stack unit **10**. The second bonding layer **40** includes a plurality of auxiliary pins **41** insulated and spaced from each other, and one auxiliary pin **41** is bonded to one primary pin **11**.

The first bonding layer **20** is located on a surface that is of the second bonding layer **40** and that is away from the primary chip stack unit **10**. The first bonding layer **20** includes a plurality of bonding components **21** insulated and spaced from each other. Each bonding component **21**

19

includes at least one bonding part **211**, and any two bonding parts **211** are insulated. Any two bonding parts **211** have an equal size in a first direction X, and each bonding part **211** in a same bonding component **21** is bonded to a same auxiliary pin **41**.

The plurality of secondary chip stack units include the second secondary chip stack unit **33** and the third secondary chip stack unit **34**. A micro bump **31** of the second secondary chip stack unit **33** is an embedded micro bump, and each micro bump **31** is bonded to each bonding part **211** in one bonding component **21** in the first bonding layer **20** by using an embedded bonding process. A micro bump **31** of the third secondary chip stack unit **34** is a solder ball, and each micro bump **31** is bonded to each bonding part **211** in one bonding component **21** in the first bonding layer **20** by using a micro bump bonding process.

Active surfaces of the second secondary chip stack unit **33** and the third secondary chip stack unit **34** face the primary chip stack unit **10**. The plurality of secondary chip stack units **30** are electrically interconnected with the primary chip stack unit **10** through active surface-to-active surface (e.g., face to face (F2F)).

As shown in FIG. 7A, a cross-sectional area of each bonding part **211** is equal to a cross-sectional area of the micro bump **31** of the second secondary chip stack unit **33**. A material of the first insulation layer **22** is a polymer material, and is the same as a material filled on a periphery of the micro bump **31** of the second secondary chip stack unit **33**.

In this way, when the micro bump **31** of the second secondary chip stack unit **33** is bonded to the bonding component **21** in the first bonding layer **20** by using the embedded bonding process, the polymer material on the periphery of the micro bump **31** is also bonded to the first insulation layer **22**. This can improve bonding reliability.

An auxiliary solder pad **41** and a micro bump **31** bonded by using a same bonding component **21** have a same cross-sectional area.

In order to add a function of the secondary chip stack unit **30** without increasing an occupied area of the secondary chip stack unit **30**, in some embodiments, as shown in FIG. 7B (a cross-sectional view of the chip package on package structure **100**), the secondary chip stack unit **30** includes a plurality of stacked secondary dies **35**, adjacent secondary dies **35** are bonded, and a secondary die **35** closest to the first bonding layer **20** is bonded to the first bonding layer **20**.

The secondary dies **35** are bonded by using wire bonding. Alternatively, the secondary die **35** may include a TSV, and TSVs in adjacent secondary dies **35** are bonded between the secondary dies **35** by using one process of micro bump bonding, embedded micro bump bonding, hybrid bonding, surface activated bonding, and atomic diffusion bonding, to implement bonding of adjacent secondary dies **35**.

In some embodiments, as shown in FIG. 7B, a manner in which adjacent secondary dies **35** are bonded is the same as a manner in which the secondary die **35** closest to the first bonding layer **20** is bonded to the first bonding layer **20**, to lower a design requirement.

To be specific, the second secondary chip stack unit **33** is bonded to the first bonding layer **20** by using an embedded micro bump bonding process, and a plurality of secondary dies **35** included in the second secondary chip stack unit **33** are also bonded by using the embedded micro bump bonding process. The third secondary chip stack unit **34** is bonded to the first bonding layer **20** by using a micro bump bonding process, and a plurality of secondary dies **35** included in the

20

third secondary chip stack unit **34** are also bonded by using the micro bump bonding process.

As shown in FIG. 7B, a second molding layer **50** covers the secondary chip stack unit **30** on a side away from the primary chip stack unit **10**, and performs molding protection for the secondary chip stack unit **30**.

A material of the second molding layer **50** may be, for example, a molding material (molding) or an insulation material.

Because a heat dissipating effect of the molding layer **50** is poor, in some embodiments, in a direction (e.g., the first direction X) perpendicular to the primary chip stack unit **10**, a surface of a secondary chip stack unit **30** of a larger size **h1** in the plurality of secondary chip stack units **30** is aligned with a surface that is of the second molding layer **50** and that is away from the primary chip stack unit **10** as shown in FIG. 7C (a cross-sectional view of the chip package on package structure **100**).

In other words, after the chip package on package structure **100** shown in FIG. 7A is manufactured, a surface that is of the secondary chip stack unit **30** and that is away from the primary chip stack unit **10** is exposed through grinding, to reduce a thickness and a thermal resistance of the second molding layer **50**.

Certainly, as shown in FIG. 7D (a cross-sectional view of the chip package on package structure **100**), when the plurality of secondary chip stack units **30** have an equal size **h1** in the first direction X, a surface that is of each secondary chip stack unit **30** and that is away from the primary chip stack unit **10** is aligned with a surface that is of the second molding layer **50** and that is away from the primary chip stack unit **10**.

In some embodiments, as shown in FIG. 8 (a cross-sectional view of the chip package on package structure **100**), the plurality of secondary chip stack units further include the first secondary chip stack unit **32**. A micro bump **31** of the first secondary chip stack unit **32** is a copper stud, and each micro bump **31** is bonded to each bonding part **211** in one bonding component **21** in the first bonding layer **20** by using a hybrid bonding process.

In this case, a cross-sectional area of the micro bump **31** of the first secondary chip stack unit **32** is less than a cross-sectional area of the micro bump **31** of the second secondary chip stack unit **33**. Therefore, a cross-sectional area of each bonding part **211** is equal to the cross-sectional area of the micro bump **31** of the first secondary chip stack unit **32**. A material of the first insulation layer **22** is a dielectric material, and is the same as a material filled on a periphery of the micro bump **31** of the first secondary chip stack unit **32**.

When the chip package on package structure **100** shown in FIG. 8 needs to be bonded to a PCB, the PCB is located on a side that is of the primary chip stack unit **10** and that is away from the first bonding layer **20**, and the first through silicon via **13** of the primary chip stack unit **10** is bonded to the PCB by using a solder ball.

In some embodiments, as shown in FIG. 9A (a cross-sectional view of the chip package on package structure **100**), the chip package on package structure **100** further includes a substrate **60**. The substrate **60** is located on a side that is of the primary chip stack unit **10** and that is away from the first bonding layer **20**, and the first through silicon via **13** of the primary chip stack unit **10** is bonded to the substrate **60**.

In other words, in this embodiment, the secondary chip stack unit **20**, the primary chip stack unit **10**, and the substrate **60** are packaged by using an embedded package

21

(embedded die in substrate package) technology. The first through silicon via 13 is directly formed in the primary die 12 to implement signal transfer. Therefore, the primary chip stack unit 10 has a small volume, and the chip package on package structure 100 has advantages such as high integration, high heat dissipating performance, and low inductance.

In some embodiments, as shown in FIG. 9B (a cross-sectional view of the chip package on package structure 100), a periphery of the micro bump 31 (e.g., a solder ball) of the third secondary chip stack unit 34 may not be filled with underfill adhesive. A gap between adjacent balls is directly filled when the second molding layer 50 is formed.

When the chip package on package structure 100 shown in FIG. 9A or FIG. 9B needs to be bonded to the PCB, a side that is of the substrate 60 and that is away from the primary chip stack unit 10 is bonded to the PCB by using a solder ball.

Embodiment 2

A difference between the embodiment 2 and the embodiment 1 lies in a structure of a primary chip stack unit 10.

As shown in FIG. 10A (a cross-sectional view of a chip package on package structure 100), the chip package on package structure 100 includes a primary chip stack unit 10, a first bonding layer 20, a plurality of secondary chip stack units 30, a second bonding layer 40, and a second molding layer 50.

The primary chip stack unit includes a primary die 12, a first redistribution layer 14, a second redistribution layer 15, and a first molding layer 16.

The first redistribution layer 14 is disposed on an active surface c1 of the primary die 12 and bonded to the primary die 12.

The second redistribution layer 15 is disposed on a non-active surface c2 of the primary die 12, and a surface that is of the second redistribution layer 15 and that is away from the primary die 12 is used as the first surface al.

The surface that is of the second redistribution layer 15 and that is away from the primary die 12 is used as the first surface al. In this case, the surface that is of the second redistribution layer 15 and that is away from the primary die 12 has a primary pin 11.

The plurality of secondary chip stack units 30 include a first secondary chip stack unit 32, a second secondary chip stack unit 33, and a third secondary chip stack unit 34. Active surfaces of the first secondary chip stack unit 32, the second secondary chip stack unit 33, and the third secondary chip stack unit 34 face the primary chip stack unit 10. The active surface c1 of the primary chip stack unit 10 faces away from the first secondary chip stack unit 32, and the non-active surface c2 of the primary chip stack unit 10 faces the first secondary chip stack unit 32. Therefore, the first secondary chip stack unit 32, the second secondary chip stack unit 33, and the third secondary chip stack unit 34 are electrically interconnected with the primary chip stack unit 10 through active surface-to-non-active surface (e.g., face to back (F2B)).

The first molding layer 16 is disposed between the first redistribution layer 14 and the second redistribution layer 15, and is located on a periphery of the primary die 12. The first molding layer 16 has a plurality of first through molding vias 161 passing through the first molding layer 16, and two ends of the first through molding via 161 are bonded to the first redistribution layer 14 and the second redistribution layer 15 respectively.

22

A material of the first molding layer 16 may be, for example, a molding material or an insulation material.

In other words, a first through silicon via 13 is not disposed inside the primary chip stack unit 10, the first through molding via 161 is disposed on the periphery of the primary chip stack unit 10, and the first molding layer 16 protects the primary chip stack unit 10 and the first through molding via 161.

Structures of the first bonding layer 20, the plurality of secondary chip stack units 30, the second bonding layer 40, and the second molding layer 50 in the embodiment 2 may be the same as those in the embodiment 1. Refer to related descriptions in the embodiment 1. Details are not described in the embodiment 2.

It should be noted that, as shown in FIG. 10A, the first secondary chip stack unit 32, the second secondary chip stack unit 33, and the third secondary chip stack unit 34 may be located directly above the first through molding via 161, and are not limited to being located directly above the primary chip stack unit 10.

In other words, the secondary chip stack unit 20, the primary chip stack unit 10, and a substrate 60 are packaged by using a fan-out wafer level package (FOWLP) technology in this embodiment. The fan-out package can increase a quantity of output pins and adjust locations of the output pins by redistributing and leading out a signal, and has a wide application scope.

When the chip package on package structure 100 shown in FIG. 10A needs to be bonded to a PCB, the PCB is located on a side that is of the first redistribution layer 14 and that is away from the first bonding layer 20, and the first redistribution layer 14 is bonded to the PCB.

As shown in FIG. 10B (a cross-sectional view of the chip package on package structure 100), the chip package on package structure 100 further includes the substrate 60, and the substrate 60 is located on a side that is of the first redistribution layer 14 and that is away from the primary chip stack unit 10. The first redistribution layer 14 is bonded to the substrate 60.

When the chip package on package structure 100 shown in FIG. 10B needs to be bonded to the PCB, the substrate 60 is bonded to the PCB.

The following describes a packaging process of the primary chip stack unit 10 shown in FIG. 10B.

As shown in FIG. 11A, first, the second redistribution layer 15 is formed on a glass carrier board.

As shown in FIG. 11B, the first through molding via 161 is formed through electroplating on the second redistribution layer 15.

As shown in FIG. 11C, the non-active surface c2 of the primary die 12 faces the second redistribution layer 15, and the primary die 12 is bonded to the second redistribution layer 15.

As shown in FIG. 11D, the periphery of the primary die 12 is filled with a molding material to form the first molding layer 16. The first molding layer 16 wraps the primary die 12 and the first through molding via 161, and exposes the first through molding via 161.

As shown in FIG. 11E, the first redistribution layer 14 is formed on the first molding layer 16. In other words, the first redistribution layer 14 is formed on the active surface c1 of the primary die 12.

As shown in FIG. 11F, the glass carrier board is removed to form the primary chip stack unit 10.

For a bonding process of the primary chip stack unit 10 and the plurality of secondary chip stack units 30, refer to the foregoing related descriptions.

23

Embodiment 3

A chip package on package structure **100** in this embodiment is an inverted multi-chip integrated structure.

As shown in FIG. **12A** (a cross-sectional view of the chip package on package structure **100**), the chip package on package structure **100** includes a primary chip stack unit **10**, a first bonding layer **20**, a plurality of secondary chip stack units **30**, a second bonding layer **40**, a substrate **60**, an underfill layer **70**, a plurality of through dielectric vias (TDVs) **80**, and a heat dissipating lid **90**.

The primary chip stack unit **10** includes a primary die **12**. An active surface of the primary die **12** is used as a first surface al.

Structures of the first bonding layer **20**, the plurality of secondary chip stack units **30**, and the second bonding layer **40** in the embodiment 3 may be the same as those in the embodiment 1. Refer to related descriptions in the embodiment 1. Details are not described in this embodiment.

An active surface of the primary chip stack unit **10** faces an active surface of the secondary chip stack unit **30**. The plurality of secondary chip stack units **30** are electrically interconnected with the primary chip stack unit **10** through active surface-to-active surface (face to face, F2F).

The underfill layer **70** is located between the primary chip stack unit **10** and the substrate **60**, and covers the plurality of secondary chip stack units **30**.

A gap between the primary chip stack unit **10** and the substrate **60** may be filled in a filling manner, for example, capillary underfill (CUF), molded underfill (MUF), non-conductive film (NCF) filling, or non-conductive paste (NCP) filling, to form the underfill layer **70**.

Because a strength of a molding material is high, in some embodiments, a molding underfill process is used to fill the molding material between the primary chip stack unit **10** and the substrate **60**, to form the underfill layer **70**.

The through dielectric via **80** of the plurality of through dielectric vias **80** is embedded in the underfill layer **70**, and two ends of the through dielectric via **80** are bonded to the substrate **60** and at least one bonding part **211** of a bonding component **21** respectively.

The through dielectric via **80** may be, for example, a copper stud. In this case, the through dielectric via **80** may be bonded to the substrate **60** through a solder ball. The through dielectric via **80** may alternatively be a solder ball or the like.

The heat dissipating lid **90** has a groove. The heat dissipating lid **90** and the substrate **60** are jointed, and the groove and the substrate **60** form a receptacle Q. The primary chip stack unit **10** and the plurality of secondary chip stack units **30** are located in the receptacle Q.

In some embodiments, as shown in FIG. **12B** (a cross-sectional view of the chip package on package structure **100**), the chip package on package structure **100** further includes a thermally conductive adhesive **95**, to improve a heat dissipating effect of the primary chip stack unit **10**. The thermally conductive adhesive **95** is located between the primary chip stack unit **10** and the heat dissipating lid **90**, and is configured to bond the primary chip stack unit **10** and the heat dissipating lid **90**.

Heat on the primary chip stack unit **10** may be directly exported through the heat dissipating lid **90** by bonding the primary chip stack unit **10** to the heat dissipating lid **90**, to improve a heat dissipating effect of the primary chip stack unit **10**.

24

When the chip package on package structure **100** needs to be bonded to a PCB, the substrate **60** is bonded to the PCB through a solder ball.

The following describes a packaging method of the chip package on package structure **100** shown in FIG. **12B**.

First, the second bonding layer **40** is formed on the primary chip stack unit **10**.

The first bonding layer **20** is formed on the second bonding layer **40**.

A third secondary chip stack unit **34**, a first secondary chip stack unit **32**, and a second secondary chip stack unit **33** are sequentially bonded to the first bonding layer **20**, to form a structure shown in FIG. **13A**.

As shown in FIG. **13B**, the through dielectric via **80** is formed on a surface that is of the first bonding layer **20** and that is away from the primary chip stack unit **10**.

As shown in FIG. **13C**, the through dielectric via **80** is bonded to the substrate **60**.

As shown in FIG. **13D**, a molding material is filled between the substrate and the first bonding layer **20** to form the underfill layer **70**.

As shown in FIG. **13E**, the thermally conductive adhesive **95** bonded to the primary chip stack unit **10** is formed on a side that is of the primary chip stack unit **10** and that is away from the substrate **60**.

The heat dissipating lid **90** and the substrate **60** are jointed, and the thermally conductive adhesive **95** and the heat dissipating lid are bonded, to form the chip package on package structure **100** shown in FIG. **12B**.

Embodiment 4

A difference between the embodiment 4 and the embodiment 3 lies in a bonding manner of a primary chip stack unit **10** and a substrate **60**.

As shown in FIG. **14** (a cross-sectional view of a chip package on package structure **100**), the chip package on package structure **100** includes a first bonding layer **20**, a plurality of secondary chip stack units **30**, a second bonding layer **40**, a substrate **60**, a third molding layer **96**, a third redistribution layer **97**, and a heat dissipating lid **90** and a thermally conductive adhesive **95**.

Structures of a primary chip stack unit **10**, the first bonding layer **20**, the plurality of secondary chip stack units **30**, the second bonding layer **40**, the heat dissipating lid **90**, and the thermally conductive adhesive **95** in the embodiment 4 may be the same as those in the embodiment 3. Refer to related descriptions in the embodiment 3. Details are not described in this embodiment.

As shown in FIG. **14**, the third molding layer **96** covers a first secondary chip stack unit **32**, a second secondary chip stack unit **33**, and a third secondary chip stack unit **34**. The third molding layer **96** has a plurality of second through molding vias **961** passing through the third molding layer **96**, and a first end of the second through molding via **961** is bonded to at least one bonding part **211** in one bonding part **21**.

The third redistribution layer **97** is disposed on a side that is of the third molding layer **96** and that is away from the primary chip stack unit **10**. The third redistribution layer **97** is bonded to a second end of the second through molding via **961**.

The substrate **60** is disposed on a side that is of the third redistribution layer **97** and that is away from the primary chip stack unit **10**, and bonded to the third redistribution layer **97**.

25

The following describes a packaging method of the chip package on package structure **100** shown in FIG. **14**.

First, the second bonding layer **40** is formed on the primary chip stack unit **10**.

The first bonding layer **20** is formed on the second bonding layer **40**.

A third secondary chip stack unit **34**, a first secondary chip stack unit **32**, and a second secondary chip stack unit **33** are sequentially bonded to the first bonding layer **20**, to form a structure shown in FIG. **13A**.

As shown in FIG. **15A**, the second through molding via **961** is formed on a surface that is of the first bonding layer **20** and that is away from the primary chip stack unit **10**.

As shown in FIG. **15B**, the second through molding via **961** is bonded to the third redistribution layer **97**.

As shown in FIG. **15C**, a molding material is filled between the first bonding layer **20** and the third redistribution layer **97** to form the third molding layer **96**.

As shown in FIG. **15D**, the third redistribution layer **97** is bonded to the substrate **60**.

As shown in FIG. **15E**, the thermally conductive adhesive **95** is formed on a side that is of the primary chip stack unit **10** and that is away from the substrate **60**.

The heat dissipating lid **90** and the substrate **60** are joined, and the thermally conductive adhesive **95** and the heat dissipating lid are bonded, to form the chip package on package structure **100** shown in FIG. **14**.

The foregoing descriptions are merely specific implementations of the present disclosure, but are not intended to limit the protection scope of the present disclosure. Any variation or replacement readily figured out by persons skilled in the art within the technical scope disclosed in the present disclosure shall fall within the protection scope of the present disclosure. Therefore, the protection scope of the present disclosure shall be subject to the protection scope of the claims.

What is claimed is:

1. A chip package on package structure, comprising:

a first chip comprising:

a first surface; and

first pins that are insulated from each other, spaced apart from each other, and disposed on the first surface;

a first bonding layer disposed on the first surface and comprising:

a second surface that is away from the first chip; and bonding components that comprise a first plurality of bonding parts and that are insulated and spaced apart from each other, wherein each bonding component of the bonding components comprises at least one bonding part of the first plurality of bonding parts, wherein each bonding part of the first plurality of bonding parts is insulated from and has a same first cross-sectional area as each other bonding part of the first plurality of bonding parts, wherein each bonding component of the bonding components is bonded to one or more respective pins of the first pins, and wherein each first pin of the first pins is bonded to at least one respective bonding part of the first plurality of bonding parts;

second chips that are stacked on the first chip, that are disposed on the second surface, and that comprise a plurality of micro bumps insulated and spaced from each other, wherein each micro bump of the plurality of micro bumps is bonded to a corresponding bonding component of the bonding components and to at least

26

one corresponding bonding part of the first plurality of bonding parts of the corresponding bonding component;

a substrate;

an underfill layer located between the first chip and the substrate and covering the second chips; and

a plurality of through dielectric vias embedded in the underfill layer, wherein two ends of each via of the plurality of through dielectric vias are respectively bonded to the substrate and a respective bonding component of the bonding components.

2. The chip package on package structure of claim 1, wherein the at least one bonding part of each bonding component of the bonding components comprises a respective second plurality of bonding parts of the first plurality of bonding parts, and wherein spacings between adjacent bonding parts in the respective second plurality of bonding parts are equal.

3. The chip package on package structure of claim 1, wherein each chip of the second chips comprises respective micro bumps of the plurality of micro bumps, wherein respective micro bumps of different chips of the second chips have different cross-sectional areas, wherein a first micro bump of the plurality of micro bumps has a smallest cross-sectional area, and wherein a second cross-sectional area of each bonding part of the first plurality of bonding parts is equal to the smallest cross-sectional area.

4. The chip package on package structure of claim 3, wherein a third chip of the second chips comprises a first bonding component of the bonding components and one or more micro bumps of the plurality of micro bumps that have the smallest cross-sectional area, and wherein a first spacing between the one or more micro bumps is equal to a second spacing between adjacent bonding parts of the bonding parts in the first bonding component.

5. The chip package on package structure of claim 1, further comprising a second bonding layer that is disposed between the first chip and the first bonding layer and that comprises a plurality of auxiliary pins insulated and spaced from each other, wherein a first end of each auxiliary pin of the plurality of auxiliary pins is bonded to a respective pin of the first pins and a second end of each auxiliary pin of the plurality of auxiliary pins is bonded to a respective bonding component of the bonding components, and wherein each auxiliary pin of the plurality of auxiliary pins is bonded to at least one respective bonding part of the first plurality of bonding parts.

6. The chip package on package structure of claim 5, wherein each auxiliary pin of the plurality of auxiliary pins that is bonded to a same bonding component of the bonding components as a corresponding micro bump of the plurality of micro bumps has a same second cross-sectional area as the corresponding micro bump.

7. The chip package on package structure of claim 1, wherein for each bonding component of the bonding components, a quantity of second bonding parts of the first plurality of bonding parts of a first bonding component of the bonding components that is bonded to a first micro bump of the plurality of micro bumps that has a second cross-sectional area is greater than a quantity of third bonding parts of the first plurality of bonding parts of a second bonding component of the bonding components that is bonded to a second micro bump of the plurality of micro bumps that has a third cross-sectional area, and wherein the second cross-sectional area is greater than the third cross-sectional area.

27

8. The chip package on package structure of claim 1, wherein the first chip comprises a first die, wherein the chip package on package structure further comprises a plurality of first through silicon vias passing through the first die, wherein an active surface of the first die is used as the first surface, and wherein the plurality of first through silicon vias are bonded to the first pins.

9. The chip package on package structure of claim 1, wherein the first chip comprises:

- a first die comprising an active surface and a non-active surface;
- a first redistribution layer disposed on the active surface and bonded to the first die;
- a second redistribution layer disposed on the non-active surface and comprising a redistribution layer surface that is away from the first die; and
- a first molding layer disposed between the first redistribution layer and the second redistribution layer, located on a periphery of the first die, and comprising a plurality of first through molding vias passing through the first molding layer, wherein two ends of each via of the plurality of first through molding vias are respectively bonded to the first redistribution layer and the second redistribution layer.

10. The chip package on package structure of claim 1, wherein the second chips comprise:

- a third chip having a first size and comprising a third surface; and
 - a fourth chip having a second size that is less than the first size,
- wherein the chip package on package structure further comprises a molding layer that covers all of the second chips on a side away from the first chip and that comprises a fourth surface that is away from the third chip, and
- wherein in a direction perpendicular to the first chip, the third surface is aligned with the fourth surface.

11. The chip package on package structure of claim 1, further comprising:

- a molding layer covering the second chips and comprising a first side that is away from the first chip;
- a plurality of through molding vias passing through the molding layer and comprising first ends and second ends, wherein each of the first ends is bonded to a respective bonding part of the bonding parts; and
- a redistribution layer that comprises a second side that is away from the first chip, that is disposed on the first side, and that is bonded to the second ends, wherein substrate is disposed on the second side and bonded to the redistribution layer.

12. The chip package on package structure of claim 1, wherein the first chip comprises a first die comprising an active surface that corresponds to the first surface, and wherein the chip package on package structure further comprises:

- a heat dissipating lid comprising a groove and coupled to the substrate, wherein the groove and the substrate form a receptacle in which the first chip and the second chips are located; and
- a thermally conductive adhesive located between the first chip and the heat dissipating lid and configured to bond the first chip and the heat dissipating lid.

13. The chip package on package structure of claim 1, wherein each of the second chips comprises a plurality of stacked dies comprising a first die that is closest to the first

28

bonding layer and bonded to the first bonding layer, and wherein adjacent dies of the stacked dies are bonded to each other.

14. An electronic device, comprising:

- a circuit board; and
- a chip package on package structure bonded to the circuit board and comprising:
 - a first chip comprising:
 - a first surface; and
 - first pins that are on the first surface and that are insulated and spaced from each other;
 - a first bonding layer disposed on the first surface and comprising:
 - a second surface that is away from the first chip; and
 - bonding components that comprise a first plurality of bonding parts and that are insulated and spaced from each other, wherein each bonding component of the bonding components comprises at least one bonding part of the first plurality of bonding parts, wherein each bonding part of the first plurality of bonding parts is insulated and has a same cross-sectional area as each other bonding part of the first plurality of bonding parts, wherein each bonding component of the bonding components is bonded to one or more respective pins of the first pins, and wherein each pin of the first pins is bonded to at least one respective bonding part of the first plurality of bonding parts;
 - second chips that are stacked on the first chip, that are disposed on the second surface, and that comprise a plurality of micro bumps insulated and spaced from each other, wherein each micro bump of the plurality of micro bumps is bonded to a corresponding bonding component of the bonding components and to at least one corresponding bonding part of the first plurality of bonding parts of the corresponding bonding component;
 - a substrate;
 - an underfill layer located between the first chip and the substrate and covering the second chips; and
 - a plurality of through dielectric vias embedded in the underfill layer, wherein two ends of each via of the plurality of through dielectric vias are respectively bonded to the substrate and a respective bonding component of the bonding components.

15. The electronic device of claim 14, wherein the at least one bonding part of each bonding component of the bonding components comprises a respective second plurality of bonding parts of the first plurality of bonding parts, and wherein spacings between adjacent bonding parts in the respective second plurality of bonding parts are equal.

16. The electronic device of claim 14, wherein the at least one bonding part of each bonding component of the bonding components comprises a respective second plurality of bonding parts of the first plurality of bonding parts, and wherein spacings between adjacent bonding parts in the respective second plurality of bonding parts are equal.

17. The electronic device of claim 14, wherein each chip of the second chips comprises respective micro bumps of the plurality of micro bumps, wherein respective micro bumps of different chips of the second chips have different cross-sectional areas, wherein a first micro bump of the plurality of micro bumps has a smallest cross-sectional area, and wherein a second cross-sectional area of each bonding part of the first plurality of bonding parts is equal to the smallest cross-sectional area.

18. The electronic device of claim 17, wherein a third chip of the second chips comprises a first bonding component of the bonding components and one or more micro bumps of the plurality of micro bumps that have the smallest cross-sectional area, and wherein a first spacing between the one or more micro bumps is equal to a second spacing between adjacent bonding parts of the bonding parts in the first bonding component. 5

19. The electronic device of claim 14, wherein the first chip comprises a first die, wherein the chip package on package structure further comprises a plurality of first through silicon vias passing through the first die, wherein an active surface of the first die is used as the first surface, and wherein the plurality of first through silicon vias are bonded to the first pins. 15

20. The electronic device of claim 14, wherein each of the second chips comprises a plurality of stacked dies comprising a first die that is closest to the first bonding layer and bonded to the first bonding layer, and wherein adjacent dies of the stacked dies are bonded to each other. 20

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